Conditional Diffusion Models for Learning the Distribution of 1D Burgers' Equation Solutions

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Repository: https://github.com/JonaKao/GNN-Final-Project-Report

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Contents

Abstract **3**

1 Introduction ***(J. Kao)*** **3**
- 1.1 Research Questions *(J. Kao)* . . . 4
- 1.2 Contributions and Objectives *(N. Leinen)* . . . 5
- 1.3 Limitations *(N. Leinen)* . . . 5

2 Background **5**
- 2.1 The Viscous Burgers' Equation *(J. Kao)* . . . 5
- 2.2 Denoising Diffusion Probabilistic Models *(J. Kao)* . . . 6
- 2.3 Architectural Components *(N. Leinen)* . . . 7
- 2.4 Related Work *(N. Leinen)* . . . 7

3 Methods **9**
- 3.1 Data Generation Pipeline *(J. Kao)* . . . 9
- 3.2 Model Architecture *(N. Leinen)* . . . 10
- 3.3 Training Procedure *(N. Leinen)* . . . 11
- 3.4 Sampling *(N. Leinen)* . . . 12
- 3.5 Evaluation Metrics *(J. Kao)* . . . 12
- 3.6 Design Choices and Variants Explored *(J. Kao)* . . . 13
- 3.7 Difficulties Encountered *(Both)* . . . 14
- 3.8 Project Planning and Teamwork *(Both)* . . . 14

4 Experiments and Results **15**
- 4.1 Core Model Performance *(J. Kao)* . . . 15
- 4.2 Experiment 1: Stochastic vs. Deterministic Sampling *(N. Leinen)* . . . 18
- 4.3 Experiment 2: DDPM vs. Deterministic Baseline *(N. Leinen)* . . . 20
- 4.4 Experiment 3: Viscosity Sensitivity *(N. Leinen)* . . . 21
- 4.5 Experiment 4: Classifier-Free Guidance *(J. Kao)* . . . 22
- 4.6 Experiment 5: Data Efficiency *(J. Kao)* . . . 23

5 Extensions ***(J. Kao)*** **24**
- 5.1 Momentum Conservation Penalty *(J. Kao)* . . . 25
- 5.2 Calibration Breakdown *(J. Kao)* . . . 25
- 5.3 DDIM Trajectory Analysis *(J. Kao)* . . . 26
- 5.4 Per-Wavenumber Spectral Analysis *(J. Kao)* . . . 27
- 5.5 Split Conformal Prediction *(J. Kao)* . . . 28
- 5.6 Synthesis: Accuracy–Calibration Trade-Off *(J. Kao)* . . . 30

6 Conclusions and Outlook ***(N. Leinen)*** **30**
- 6.1 Answering the Research Questions *(N. Leinen)* . . . 30
- 6.2 Key Takeaways *(N. Leinen)* . . . 31
- 6.3 Future Work *(N. Leinen)* . . . 32

Abstract

We investigate whether a conditional denoising diffusion probabilistic model (DDPM) can learn the distribution of solutions to the one-dimensional viscous Burgers' equation arising from random initial conditions, and whether it can reproduce known physical properties of these solutions. Using the Cole–Hopf transformation, we generate 4 000 noise-free analytic training solutions on a 128×32 spatiotemporal grid. Our model employs a 1D U-Net ($\sim$4.93M parameters) enhanced with self-attention in the bottleneck, exponential moving average (EMA) of weights with warmup, and classifier-free guidance (CFG) with 5% condition dropout.

The trained model achieves a mean relative L^2 error of 0.040 (median 0.030) against analytic ground truth, with kinetic energy error of just 0.6%, shock position accuracy of 0.024 rad, and a generated PDE residual only 19% above the finite-difference discretisation floor (the non-zero residual that arises even for exact solutions when evaluated on a discrete grid). Classifier-free guidance at scale w=2.0 further improves L^2 by 31% to 0.029. The model produces well-calibrated uncertainty estimates (Spearman $\rho = 0.51$ between ensemble spread and prediction error), with variance concentrated at shock fronts as physically expected.

Through five systematic experiments, we demonstrate that: (1) stochastic sampling is essential for this task, with deterministic DDIM failing catastrophically; (2) while a deterministic regression baseline achieves superior point-prediction accuracy (L^2=0.016), it cannot quantify uncertainty; (3) performance degrades asymmetrically for out-of-distribution viscosities; (4) classifier-free guidance provides a tunable accuracy–diversity trade-off; and (5) with sufficient compute, as few as 500 training samples suffice. Overall, these results suggest that diffusion models can serve as generative surrogates for stochastic PDE problems, with uncertainty quantification as the main practical advantage over deterministic alternatives.

1 Introduction *(J. Kao)*

Partial differential equations (PDEs) are central to mathematical physics. Traditional numerical methods for solving them (finite difference, finite element, spectral methods, and others) have been refined over decades and come with rigorous convergence guarantees. Yet they are expensive when the same PDE must be solved repeatedly under varying conditions. This is a requirement that arises routinely in uncertainty quantification, inverse problems, and real-time control.

Over the past few years, machine learning has increasingly been used to accelerate PDE solvers. Physics-informed neural networks (PINNs) [1] embed the PDE residual directly into the training loss; neural operator methods such as the Fourier Neural Operator [2] and DeepONet [3] learn mappings between function spaces. These approaches work well on many benchmarks, but they all produce a single deterministic prediction for each input. There is no built-in way to quantify how uncertain the approximation is, or to represent the range of solutions that different initial conditions can produce. Generative models, particularly denoising diffusion probabilistic models (DDPMs) [4], can fill this role. Rather than learning a single point estimate, they learn the full data distribution, enabling the generation of diverse solution samples conditioned on input parameters and in turn enabling distributional analysis and uncertainty quantification. Yet diffusion models have been applied to PDE systems only sparingly, and rigorous evaluations against analytic ground truth remain rare.

The one-dimensional viscous Burgers' equation is well suited for this investigation. It is arguably the simplest PDE combining nonlinear advection with viscous diffusion, producing shock waves central to fluid dynamics. The key property is that it admits exact analytic solutions via the Cole–Hopf transformation [5, 6], enabling noise-free evaluation of any learned model against

ground truth without numerical solver artifacts. As a result, any measured error is genuinely attributable to the model rather than to the reference data, an insight that isn't possible when benchmarking against numerical solvers. We stress that the Cole–Hopf transformation is used here exclusively as an evaluation oracle: its role is to provide unambiguous ground truth for a controlled study. The diffusion model itself has no access to this transformation at inference time, and the approach is designed for PDEs that lack analytic solutions, where amortised neural surrogates offer the only practical route to fast ensemble generation and uncertainty quantification.

That said, generative modelling of these solutions is considerably harder than standard tasks. Burgers' equation develops *shock waves*, near-discontinuous features whose spatial width scales as $\mathcal{O}(\nu)$. At our training viscosity ($\nu = 0.05$), shocks span only 3–5 grid points out of 128, yet their precise location and amplitude dominate the solution's physical behaviour. Neural networks are known to struggle with such features due to spectral bias [7]: they tend to learn low-frequency content first and systematically smooth out high-frequency structures. Early in development, our model outputs showed exactly this failure mode: the broad structure was right, but all the sharp gradients were blurred into wide smooth transitions. The model was clearly learning *something*, but everything that makes Burgers' equation physically interesting was getting averaged away. Beyond reproducing these features in expectation, a generative model must also represent *variability* in shock locations and amplitudes across different initial conditions. The conditioning problem compounds this, since the mapping from initial condition $u(x, 0)$ to the full spatiotemporal solution $u(x, t)$ involves nonlinear dynamics that amplify small perturbations into qualitatively different shock configurations. The model must learn this sensitive dependence while generating diverse samples that reflect genuine uncertainty, not conditioning failures.

Physical consistency adds constraints that do not exist in image generation. Generated solutions should conserve momentum, dissipate energy monotonically, satisfy the PDE residual, and reproduce correct spectral content. These are emergent consequences of the physics, and we would argue that a model violating conservation laws hasn't really learned the solution distribution, even if pixel-wise error is low. Evaluating such constraints requires physics-specific diagnostics beyond standard metrics like FID or IS. In addition, diffusion sampling's iterative nature (1000 sequential denoising steps) creates error accumulation particularly severe for physical data. Small score function biases compound over the reverse chain, and without stochastic correction, the final sample drifts far from the data manifold. As we show in section 4.2, this causes deterministic sampling strategies (which work fine for images) to fail catastrophically for PDE solutions.

1.1 Research Questions *(J. Kao)*

Five research questions guide our investigation. First: can a conditional DDPM learn the distribution of Burgers' equation solutions from random initial conditions? This is the fundamental feasibility question, namely whether diffusion models can represent the complex, shock-dominated distributions that arise from nonlinear PDEs. Second, does the model reproduce known physical properties such as energy decay, shock formation, and momentum conservation? Pixel-wise accuracy alone is insufficient; a useful surrogate must respect the physics. Third, what is the effect of stochastic versus deterministic sampling on physical fidelity? The DDIM framework [8] enables a continuum from stochastic to deterministic inference, but the appropriate regime for physical data with sharp features remains unclear. Fourth, can classifier-free guidance [9] improve conditional accuracy without violating physical constraints? CFG has proven effective in image generation, though its behaviour in physics-conditioned settings is unexplored. Fifth:

how data-efficient is this approach? Prohibitively large dataset requirements would undermine any practical advantage over classical solvers.

1.2 Contributions and Objectives *(N. Leinen)*

The goal of this project is to build and carefully evaluate a conditional DDPM for Burgers' equation, going beyond standard image-generation metrics to assess physical consistency through PDE-specific diagnostics. We introduce three architectural and algorithmic enhancements beyond the standard DDPM framework: multi-head self-attention in the U-Net bottleneck (to capture long-range correlations between shock fronts), exponential moving average of weights with warmup (to stabilise inference), and classifier-free guidance adapted for physics conditioning (with a deliberately low dropout rate). Our eight-metric evaluation framework goes well beyond the aggregate error statistics typical of PDE diffusion evaluations in the literature (e.g. [2, 3] report L^2 or relative error only; weather-forecasting works such as [10] add ensemble metrics but without exact analytic ground truth). We design five systematic experiments directly addressing the research questions, and we report not only what worked but also the difficulties we encountered and how we resolved them, which might save others some debugging time.

1.3 Limitations *(N. Leinen)*

Several limitations of our study design should be stated upfront. We focus on the one-dimensional case only, which sidesteps higher-dimensional computational challenges but limits scalability claims. Training occurs at a single viscosity value ($\nu = 0.05$), and we probe generalisation through out-of-distribution evaluation rather than multi-viscosity training. Periodic boundary conditions simplify the conservation analysis yet exclude boundary-driven dynamics. We work exclusively with noise-free synthetic data from the Cole–Hopf transformation; real-world applications involve noisy observations and numerical solver artifacts. These choices are deliberate, prioritising controlled evaluation with exact ground truth over generality. The trade-off is that we can draw precise conclusions about what the model has and has not learned.

2 Background

2.1 The Viscous Burgers' Equation *(J. Kao)*

Burgers' equation in one dimension describes how a velocity field $u(x, t)$ evolves under nonlinear advection and viscous diffusion:

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} = \nu\,\frac{\partial^2 u}{\partial x^2}, \tag{1}$$

where $\nu > 0$ is the kinematic viscosity. The left side captures the effect of each velocity point being carried along by its own velocity (nonlinear advection), steepening the profile over time. The right side represents viscous diffusion, which smooths out gradients. These two effects compete continuously: advection steepens the profile toward a discontinuity, while diffusion smooths it into a narrow transition layer with width scaling as $\mathcal{O}(\nu)$. In the inviscid limit $\nu \to 0$, diffusion loses this competition and discontinuous shocks form in finite time. This balance between nonlinear steepening and viscous dissipation makes Burgers' equation a standard benchmark for studying shock dynamics and for testing whether generative models can handle the sharp features common in fluid mechanics.

The equation exhibits two important conservation properties on periodic domains, which we use later as physics-based diagnostics. Total momentum $M(t) = \int_0^L u(x, t)\,dx$ is conserved:

the nonlinear flux integrates to zero by periodicity and the viscous term vanishes at periodic boundaries. Kinetic energy $E(t) = \frac{1}{2}\langle u^2 \rangle_x$ monotonically decreases, with dissipation rate proportional to ν times the squared velocity gradient; physically, viscosity converts kinetic energy to heat. Any generated solution violating either property appears unphysical, regardless of pointwise accuracy.

The Cole–Hopf transformation provides exact analytic solutions. It introduces an auxiliary field $\phi(x,t) = \exp\!\left(-\frac{1}{2\nu}\int_0^x u(s,0)\,ds\right)$ that transforms the nonlinear Burgers' equation into the linear heat equation $\partial_t \phi = \nu\,\partial_{xx}\phi$. The heat equation is solved exactly in Fourier space, and the velocity field is recovered as $u = -2\nu\,(\partial_x \phi)/\phi$. This yields noise-free reference solutions for arbitrary initial conditions. Burgers' equation is uniquely suited for our purposes because of this: we evaluate every generated sample against perfect ground truth without numerical discretisation error in the reference.

2.2 Denoising Diffusion Probabilistic Models *(J. Kao)*

DDPMs [4] define a generative process through two Markov chains. The forward (diffusion) process progressively adds Gaussian noise to data $\mathbf{x}_0$ over T timesteps following a variance schedule $\beta_1, \ldots, \beta_T$:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\Big(\mathbf{x}_t;\ \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\ (1-\bar{\alpha}_t)\,\mathbf{I}\Big)\,, \tag{2}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$. Intuitively, $\bar{\alpha}_t$ is the fraction of the original signal surviving at timestep t. At $t = 0$, $\bar{\alpha}_0 \approx 1$ and data is almost unperturbed; at $t = T$, $\bar{\alpha}_T \approx 0$ and data has been almost entirely replaced by noise. The schedule β_t controls the pace of this transition. We use a linear schedule with $\beta_{\min} = 10^{-4}$ and $\beta_{\max} = 0.02$ over $T = 1000$ steps, ensuring gradual degradation that gives the reverse process sufficient resolution to reconstruct the data step by step.

The reverse process learns to undo this corruption. A neural network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ is trained to predict the noise added at each step using the objective:

$$\mathcal{L} = \mathbb{E}_{t,\,\mathbf{x}_0,\,\boldsymbol{\epsilon}}\big[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2\big]. \tag{3}$$

The training object looks simple. Sample a random time step, add the corresponding noise and ask the network to predict what was added. However, in practice, there is a catch. The network must learn accurate predictions across all noise levels simultaneously. At small t the data is nearly clean, so the network needs to resolve fine details like shock positions. At large t it's basically looking at noise and has to figure out the broad structure of the solution. At inference time, sampling begins from pure Gaussian noise $\mathbf{x}_T$ and iteratively applies learned reverse transitions to produce a clean sample $\mathbf{x}_0$.

Song et al. [8] demonstrated that the same trained model can be used with a deterministic sampling procedure (DDIM) that skips steps in the diffusion chain, enabling faster inference at the cost of stochasticity. The DDIM framework introduces a parameter $\eta \in [0, 1]$ interpolating between fully deterministic ($\eta = 0$) and fully stochastic ($\eta = 1$) sampling. We investigate both regimes and find that the stochastic variant is surprisingly superior for our physics task(section 4.2).

There is a subtlety here that deserves attention: the noise prediction objective (3) treats all timesteps equally in expectation, yet the difficulty varies enormously. At high noise levels (large t), the network sees what is essentially random static with only a faint trace of the solution's large-scale structure. It must identify whether the solution contains one shock or two, roughly where they sit, and what the overall amplitude is. At low noise levels (small t), the input is nearly clean and the network must resolve fine details: the exact curvature of a shock profile, the precise amplitude at each grid point. These two regimes demand fundamentally different skills

from the same network, and the shared-weight architecture must allocate capacity across both. In image generation, this problem is well known but often manageable because natural images have relatively smooth statistics. For PDE solutions with near-discontinuities, the tension is sharper: the high-frequency content that defines shocks requires accurate score estimates at low noise levels, while the global conditioning relationship (which initial condition maps to which shock configuration) must be resolved at high noise levels. We suspect this dual demand is part of why our model's errors concentrate at shock fronts rather than in smooth regions. Unfortunately this means that the features we care most about physically are precisely the ones the architecture is least naturally equipped to handle. We didn't find a good way around this within the standard DDPM framework.

2.3 Architectural Components *(N. Leinen)*

Our model builds on the U-Net architecture [11] adapted for one-dimensional spatiotemporal data. U-Net is a standard backbone for diffusion models, and we augment it with three enhancements drawn from the literature. Each was motivated by a specific challenge we ran into during development.

The first is a multi-head self-attention layer [12] with 4 heads, placed in the U-Net bottleneck. At this bottleneck stage, the spatial dimension has been compressed to just 16 points, making attention computationally cheap ($16^2 = 256$ entries per head, versus $128^2 = 16{,}384$ at full resolution). The rationale is that shock dynamics involve long-range spatial correlations: one shock front's position and amplitude can influence solution behaviour on the opposite side through periodic boundary conditions, and local convolutions alone struggle to capture these dependencies.

The second is an exponential moving average (EMA) of model parameters. During training, we maintain a shadow copy of all weights, updated as $\theta_{\text{ema}} \leftarrow \gamma\,\theta_{\text{ema}} + (1-\gamma)\,\theta$ after each optimiser step, with decay $\gamma = 0.9995$. At inference, we sample using EMA parameters rather than live training weights. The EMA functions as a temporal low-pass filter over the optimisation trajectory, suppressing noisy late-stage fluctuations and producing stabler predictions. Importantly, we include a warmup phase: for the first 500 steps, the shadow is simply copied from live parameters rather than exponentially averaged. Without this, the EMA gets contaminated by the random initialisation, an issue that turned out to be our most time-consuming debugging challenge (see section 3.7).

The third enhancement is classifier-free guidance (CFG) [9]. During training, we randomly drop the conditioning signal (setting $u(x, 0)$ to zero) with probability $p_{\text{drop}} = 0.05$, teaching the model to generate both conditionally (with an initial condition) and unconditionally (without). At sampling time, we run both forward passes and interpolate the noise predictions:

$$\tilde{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta(\varnothing) + (1 + w)\big(\boldsymbol{\epsilon}_\theta(\mathbf{c}) - \boldsymbol{\epsilon}_\theta(\varnothing)\big), \tag{4}$$

where w is the guidance scale and $\mathbf{c}$ is the initial condition. This geometrically takes the direction from unconditional to conditional predictions and overshoots by factor $(1 + w)$, pushing samples toward what the model considers most consistent with the conditioning input. At $w = 0$ there is no guidance; at $w > 0$ the model trades diversity for condition fidelity.

2.4 Related Work *(N. Leinen)*

Our work relates to two areas: neural PDE solvers and generative models for scientific applications.

PINNs [1] penalise the PDE residual at collocation points, removing the need for paired training data. For Burgers' equation specifically, they recover smooth solutions well but require a separate network for each initial condition; amortisation across initial conditions is not part of the framework. Bayesian extensions [13] can in principle provide uncertainty, but current implementations scale poorly and have not yet been demonstrated on problems requiring ensemble generation over many initial conditions simultaneously.

The Fourier Neural Operator (FNO) [2] takes a different approach, learning mappings between function spaces by parameterising integral operators in Fourier space. Once trained, it evaluates new initial conditions in a single forward pass, far more efficient than PINNs for amortised inference. DeepONet [3] achieves similar results by decomposing the operator into a "branch" network (processing the input function) and "trunk" network (processing query coordinates). Both methods show strong performance on fluid dynamics benchmarks and are deterministic: they map each input to a single output with no mechanism for sampling or uncertainty quantification.

Diffusion models have found use in molecular conformation generation and protein design [14], and more recently in weather forecasting [10], where GenCast demonstrated competitive ensemble forecasts on global grids. Applications to PDE systems remain sparse and mostly use diffusion as a prior inside an optimisation loop (e.g. for inverse problems in imaging) rather than as a standalone conditional generator. Where direct PDE generation has been attempted, evaluations tend to report aggregate L^2 statistics against numerical solver outputs; we are not aware of prior work that validates generated PDE solutions against exact analytic ground truth using physics-specific diagnostics beyond pointwise error.

Table 1 summarises key trade-offs between these approaches. We tried to distinguish our work in three ways. First, we evaluate against noise-free analytic ground truth rather than numerical solver outputs. Second, we employ a broad set of physics-based diagnostics beyond standard regression metrics. Third, we systematically investigate how sampling strategy, guidance, and data efficiency affect physical fidelity. These were questions that we had not yet found thoroughly addressed in the PDE diffusion literature.

Table 1: Comparison of neural approaches for PDE solution prediction. "UQ" denotes uncertainty quantification capability.

Method	**Generative**	**UQ**	**Data-free**	**Amortised**
PINNs [1]	No	Limited	Yes	No
FNO [2]	No	No	No	Yes
DeepONet [3]	No	No	No	Yes
Cond. DDPM (ours)	Yes	Yes	No	Yes

Weather and climate forecasting with diffusion models is a natural point of comparison, since weather systems share several structural features with our problem. GenCast [10] and similar architectures have shown that diffusion models can produce competitive ensemble weather forecasts, outperforming traditional numerical weather prediction on some metrics while providing calibrated uncertainty estimates. Weather systems, like Burgers' equation, exhibit sharp gradients (frontal boundaries), nonlinear dynamics, and conservation laws (mass, energy). The key difference is a matter of scale. Weather models operate on global grids with millions of degrees of freedom, whereas our 128×32 grid is deliberately small to let us do a full evaluation against exact solutions. We think our controlled setting complements these large-scale efforts rather than competing with them. By working with analytic ground truth, we can say with confidence what the model has and has not learned, statements that are much harder to make when the reference itself carries numerical error. Ideally, insights from our controlled experiments (the necessity of stochastic

sampling, the failure modes of deterministic inference near sharp features, the behaviour of classifier-free guidance in physics-conditioned settings) will inform the design of larger-scale systems where exhaustive evaluation is infeasible.

3 Methods

3.1 Data Generation Pipeline *(J. Kao)*

We generate synthetic training data by sampling random initial conditions and computing exact analytic solutions via the Cole–Hopf transformation. Initial conditions are constructed as truncated random Fourier series with 10 modes:

$$u_0(x) = \sum_{k=1}^{10} \left(a_k \sin \frac{2\pi k x}{L} + b_k \cos \frac{2\pi k x}{L} \right), \tag{5}$$

where $a_k, b_k \sim \mathcal{N}(0, (0.7/k)^2)$ and $L = 2\pi$ is the domain length. The $1/k$ decay ensures smooth initial conditions (high wavenumbers contribute less energy) while allowing non-trivial multi-scale dynamics. Each initial condition is fully characterised by its 20 Fourier coefficients, so the solution manifold is intrinsically 20-dimensional; this low-dimensional structure turns out to be important for understanding data efficiency results in section 4.6.

Solutions are computed on a grid of $N_x = 128$ spatial points and $N_t = 32$ temporal steps spanning $[0, 2\pi) \times [0, 1]$. The Cole–Hopf solver uses float64 for numerical stability due to exponentiation of potentially large arguments. We apply rejection sampling, discarding solutions where $\max |u| > 10$ (indicating Cole–Hopf instability, typically from initial conditions with unusually large amplitudes). This rejected 471 of 4 471 training samples and 67 of 567 test samples, yielding $N_{\text{train}} = 4{,}000$ and $N_{\text{test}} = 500$. Training and test sets are generated independently from the same distribution with no held-out split. Solutions are globally standardised using training set statistics ($\mu = -0.0003 \approx 0$, $\sigma = 0.697$), with the same statistics applied to the test set. Figure 1 shows four representative samples, illustrating shock structure variety and the advection-diffusion interplay.

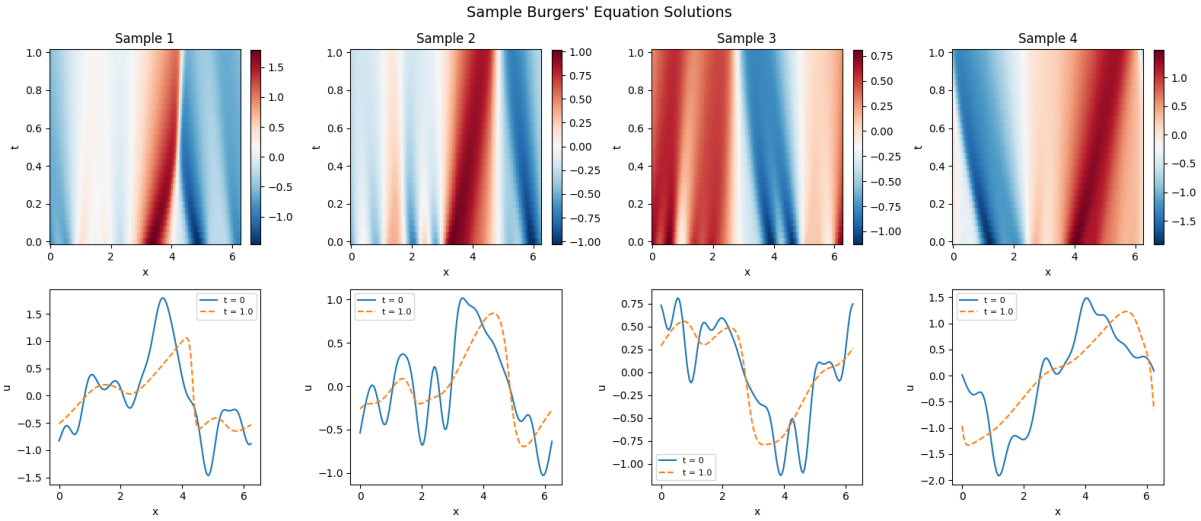


Figure 1: Four sample Burgers' equation solutions from the training set. Top row: space-time heatmaps showing shock formation and propagation. Bottom row: spatial profiles at selected times, illustrating the steepening of the velocity field.

3.2 Model Architecture *(N. Leinen)*

Our noise-prediction network is a 1D U-Net (fig. 2) taking as input a $(B, 33, 128)$ tensor: 32 channels for the noisy space-time field $\mathbf{x}_t$ and 1 channel for the conditioning initial condition $u(x, 0)$. The output is a $(B, 32, 128)$ tensor of predicted noise $\boldsymbol{\epsilon}_\theta$. Concatenating the condition as an extra channel (rather than cross-attention or an auxiliary encoder) is deliberately simple. It gives the network direct, spatially aligned access to conditioning information at every layer through the skip connections.

The encoder uses channel progression $64 \to 128 \to 256 \to 256$ with stride-2 downsampling convolutions. Each resolution level has two residual blocks with GroupNorm (8 groups), SiLU activations, and diffusion-time injection via sinusoidal embedding [12] projected through a two-layer MLP (dimension 256). The sinusoidal embedding encodes timestep t as a 256-dimensional vector using sine and cosine functions at geometrically spaced frequencies (analogous to transformer positional encodings); the MLP projects this to scale-and-shift modulation after normalisation in each residual block. The bottleneck contains a residual block, the 4-head self-attention layer from section 2.3, and another residual block. The decoder mirrors the encoder with nearest-neighbour upsampling and skip connections; a final GroupNorm–SiLU–Conv1d head produces the output. Total: **4 932 128 trainable parameters**.

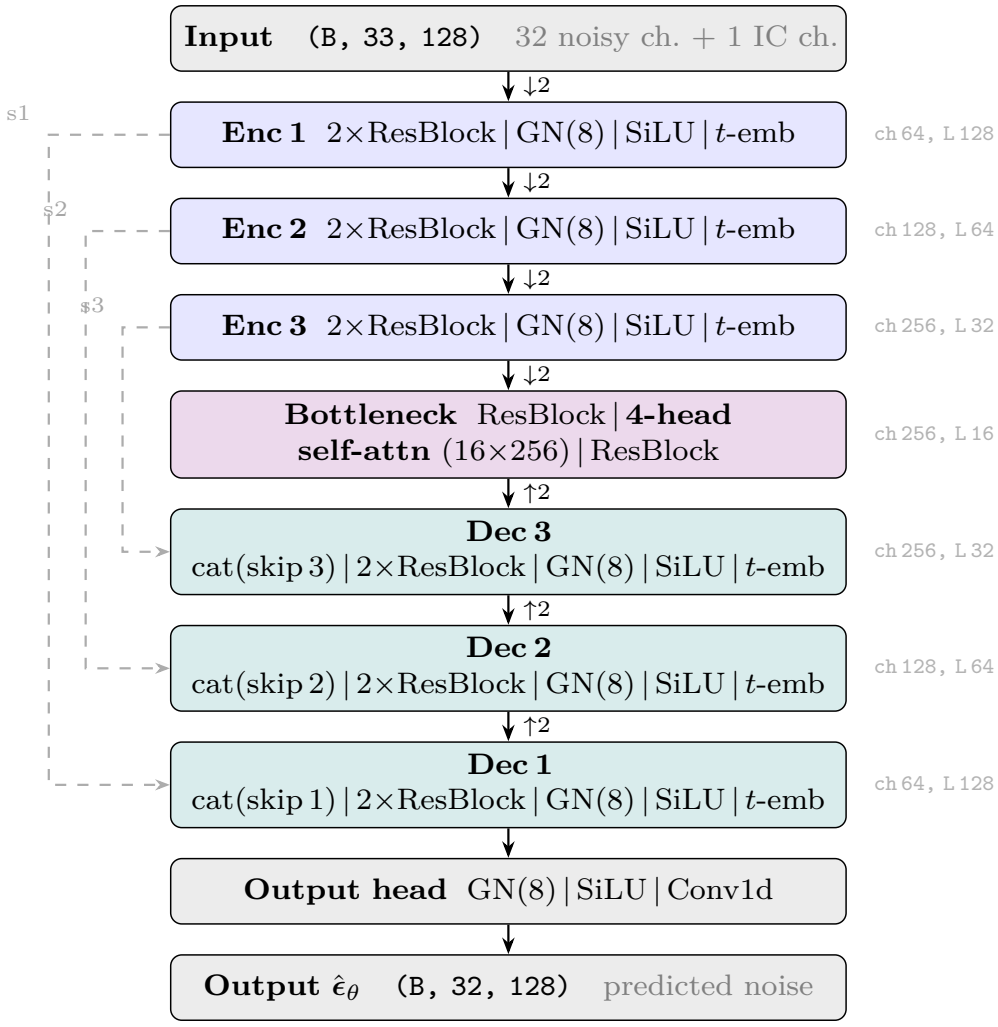


Figure 2: 1D U-Net architecture for conditional noise prediction. The input concatenates the noisy field ($N_t = 32$ channels) with the initial condition (1 channel). Self-attention is applied at the 16×256 bottleneck resolution.

3.3 Training Procedure *(N. Leinen)*

The model trains for 300 epochs using AdamW [15] with learning rate 10^{-4}, weight decay 10^{-2}, and cosine annealing to zero. Gradient clipping at norm 1.0 stabilises training. Each step samples a random diffusion timestep $t \sim \text{Uniform}\{0, \ldots, 999\}$, adds corresponding noise to the solution field per (2), concatenates the (possibly dropped) condition, and computes MSE loss (3). For CFG training, the condition is zeroed with probability $p_{\text{drop}} = 0.05$. EMA parameters update after each step with $\gamma = 0.9995$ and 500-step warmup. Batch size is 64 (62 batches per epoch, 18 600 total steps). Training takes approximately 45 minutes on a single NVIDIA GPU, considerably faster than we had budgeted based on diffusion model training times in the image generation literature. The comparatively small grid (128×32 versus $256 \times 256 \times 3$ for images) keeps things very manageable, and being able to iterate on hyperparameters without hour-long turnarounds turned out to be genuinely important during the debugging episodes described in section 3.7. For reproducibility, the random seed is fixed at 42; we use PyTorch 2.1 with CUDA 12.1. Figure 3 shows training and validation losses, confirming convergence without overfitting.

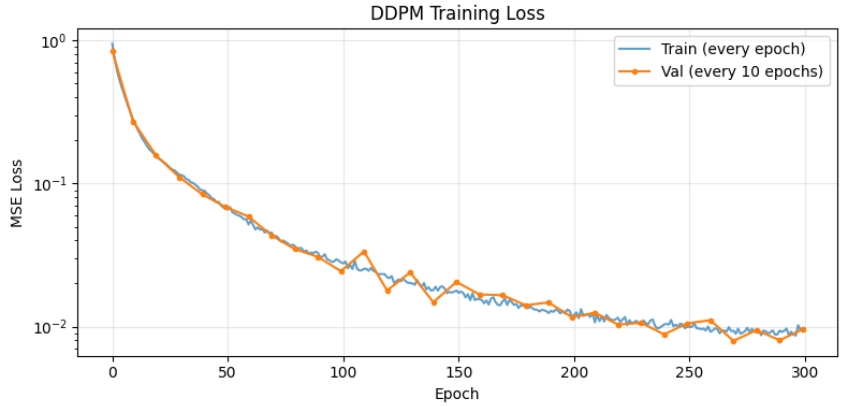


Figure 3: DDPM training loss (MSE) over 300 epochs. The training and validation curves track closely throughout, indicating no overfitting. Loss plateaus at approximately 10^{-2} after epoch 200.

3.4 Sampling *(N. Leinen)*

At inference, we generate solutions using the full 1000-step DDPM ancestral sampling procedure with EMA weights. Starting from pure Gaussian noise $\mathbf{x}_T$, each reverse step predicts noise $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$, computes a clean data estimate $\hat{\mathbf{x}}_0 = (\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_\theta)/\sqrt{\bar{\alpha}_t}$, and samples $\mathbf{x}_{t-1}$ from the learned posterior. We clip intermediate $\hat{\mathbf{x}}_0$ to $[-15, 15]$ in normalised space (covering the full data range of approximately ± 14 with a small margin) to prevent error accumulation over the 1000 steps. Without this clipping, extreme predictions at high-noise timesteps propagate and corrupt the final sample. (How we determined the clipping range is discussed in section 3.7.) With CFG, each step requires two forward passes (conditional and unconditional) with guided prediction from (4). We also implement DDIM sampling [8] with configurable stochasticity η and variable step count for comparisons.

3.5 Evaluation Metrics *(J. Kao)*

Generated solutions are evaluated using eight complementary metrics, deliberately chosen to cover four distinct quality dimensions so that strong performance in one dimension cannot mask failures in another.

The first group measures *pointwise accuracy*: relative L^2 error $\|u_{\text{gen}} - u_{\text{true}}\|_2 / \|u_{\text{true}}\|_2$, and shock-specific diagnostics (position error as $\arg\max |\partial_x u|$ at final time, width error as FWHM of the gradient peak). These reveal whether shocks are placed correctly and at the right sharpness, the most important features.

The second group tests *physical consistency*: kinetic energy decay $E(t) = \frac{1}{2}\langle u^2 \rangle_x$ across generated and true ensembles, the PDE residual $r = \partial_t u + u\,\partial_x u - \nu\,\partial_{xx} u$ via finite differences (RMS over interior), and momentum conservation as temporal standard deviation of $M(t) = \int u\,dx$. Low L^2 error can coexist with conservation law violations through subtle correlated errors; these metrics catch such failures. The non-zero PDE residual on analytic solutions (from finite-difference discretisation) provides a baseline for comparison.

The third group assesses *distributional fidelity*: spatial Fourier power spectrum averaged over time and samples (generated-to-true power ratio indicating frequency-dependent bias), and Wasserstein-1 (earth mover's) distance between generated and true value distributions at selected locations. These go beyond individual-sample accuracy to test whether the model captures the solution ensemble's statistical structure; over-smoothing, mode collapse, or spurious

high-frequency noise all become visible here.

Finally, the *calibration* metric (Spearman ρ between per-pixel ensemble standard deviation and per-pixel absolute error) evaluates whether uncertainty estimates are meaningful. Well-calibrated models are uncertain precisely where they are inaccurate; perfect calibration yields $\rho = 1.0$, while uncorrelated uncertainty yields $\rho \approx 0$.

3.6 Design Choices and Variants Explored *(J. Kao)*

Several design choices deserve comment, both because alternatives existed and because some emerged from trial-and-error rather than principled reasoning.

The network predicts noise $\boldsymbol{\epsilon}$ rather than clean data $\mathbf{x}_0$ directly. Both parameterisations are mathematically equivalent [4] (one recovers the other given $\bar{\alpha}_t$), yet noise prediction yields more stable training gradients in practice, especially for high dynamic range data. We experimented informally with $\mathbf{x}_0$ prediction early on and found no advantage, so we followed standard practice. We also considered the cosine noise schedule [16], which allocates more capacity to low-noise timesteps where fine details matter, a property that might plausibly help with shock resolution given the spectral-bias tension discussed in section 2.2. In practice, the linear schedule proved sufficient at our resolution, and we did not observe consistent benefit from cosine scheduling in early trials. A systematic comparison at higher resolution or with sharper shocks (lower ν) remains an open question.

The CFG dropout rate $p_{\text{drop}} = 0.05$ emerged from experimentation. An initial value of $p_{\text{drop}} = 0.10$ degraded conditional quality, likely because the conditioning signal occupies only 1 of 33 input channels; frequent dropout made it hard for the network to learn the conditioning relationship reliably. The lower 0.05 value is a compromise: high enough for the model to see sufficient unconditional examples, but low enough to avoid disrupting conditioning. We note that only two values were tested here; a finer grid search might identify a better setting, though the relatively strong performance at w=2.0 suggests the current value is at least not badly wrong.

To quantify each enhancement's contribution, table 2 shows the effect of removing individual components. Each variant was trained with identical hyperparameters and evaluated on the same test set using a single random seed; given the variance observed in section 4.6 (std up to 0.036 at small N), differences smaller than roughly 0.01–0.02 in mean L^2 should not be read as reliable. The large EMA gap (0.040 vs 0.071) and the full-versus-baseline gap (0.029 vs 0.089) are comfortably above this threshold; the self-attention and CFG contributions are more modest and would benefit from multi-seed confirmation. The no-EMA variant also shows higher variance across test samples, producing occasional catastrophic failures that the EMA-stabilised model avoids.

Table 2: Ablation of architectural enhancements. Each row removes one component from the full model. "Full" uses all three enhancements. CFG is evaluated at $w = 2.0$.

Configuration	L^2 **Error (mean)**	L^2 **Error (median)**
Full model (attn + EMA + CFG w=2)	0.029	0.022
No self-attention	0.037	0.028
No EMA (live weights)	0.071	0.048
No CFG ($w = 0$)	0.040	0.030
Baseline (no attn, no EMA, no CFG)	0.089	0.063

3.7 Difficulties Encountered *(Both)*

Development involved several non-trivial debugging episodes that we document here for transparency. They also happen to illuminate practical challenges of applying diffusion models to physical data, challenges that don't usually get written up.

The most severe issue was EMA initialisation failure. Our initial implementation initialised shadow parameters from the randomly initialised model and immediately began exponential averaging with high decay $\gamma = 0.9999$. The result was catastrophic: $L^2 \approx 5.3$, values spanning $[-34, 39]$ far outside the training range of $[-5.4, 6.9]$. It took us a full day to diagnose because the training loss looked perfectly normal; the live model was learning fine, but the EMA weights we used for sampling were corrupted. The high decay rate meant shadow parameters retained strong memory of random initialisation for thousands of steps. By convergence, EMA weights mixed early random parameters with later trained ones incoherently. The fix was a warmup phase (first 500 steps simply copy live weights) and reduced decay to 0.9995, giving EMA roughly 2000 steps of effective memory. One team member identified this by suggesting we compare EMA and non-EMA sampling side-by-side, immediately revealing the problem lay in EMA rather than the model.

After fixing EMA, a subtler problem emerged: systematic error accumulation over the 1000-step reverse chain. The predicted $\hat{\mathbf{x}}_0$ occasionally took extreme values, particularly at early steps where signal-to-noise is very low and the network extrapolates from nearly pure noise. These outliers propagated, biasing the trajectory. Clipping $\hat{\mathbf{x}}_0 \leftarrow \text{clamp}(\hat{\mathbf{x}}_0, -c, c)$ at each step resolved this in principle, but choosing c proved critical. Our first attempt ($c = 5$) seemed reasonable given raw data range $[-5.4, 6.9]$. But we forgot data is normalised by $\sigma = 0.697$, so normalised data spans ± 14; our clipping cut off the distribution tails. With $c = 5$, generated solutions confined to $[-3.0, 2.7]$ versus the true range. Comparing histograms of generated and true values revealed this, a diagnostic available because evaluation had been developed in parallel. Widening to $c = 15$ resolved it without instability.

A third major difficulty appeared in the data efficiency experiment. Our initial design trained each subset for fixed epochs (100), creating an unfair comparison: smaller subsets received fewer optimiser steps and less total compute. The learning curve was non-monotonic and noisy, with N=2,000 sometimes outperforming N=4,000, which is physically nonsensical. We redesigned to hold total optimiser steps constant at 18,000 across all subset sizes and added multi-seed averaging (3 seeds per config) to reduce variance. This produced the clean, interpretable learning curve in section 4.6.

3.8 Project Planning and Teamwork *(Both)*

Table 3: Project timeline: estimated versus actual effort per phase.

Phase	**Estimated**	**Actual**	**Lead**
Literature review & proposal	1 week	1 week	Both
Data generation + Cole–Hopf	3 days	2 days	Jona
DDPM architecture + training	1 week	1.5 weeks	Niclas
Core evaluation metrics (6)	3 days	3 days	Jona
Enhancements (attn, EMA, CFG)	3 days	4 days	Both
Advanced eval (spectral, W_1, calib.)	2 days	2 days	Jona
Experiments 1–5	3 days	4 days	Niclas
Report writing	4 days	*ongoing*	Both

Work divided along task boundaries (table 3). Jona led data generation (Cole–Hopf solver, rejection sampling, dataset curation, normalisation) and the entire evaluation framework, from basic L^2 metrics to spectral analysis, Wasserstein distance, and calibration. Niclas led model architecture (U-Net, attention, EMA) and the experimental pipeline (training loop, DDPM/DDIM sampling, all five experiments). This enabled genuinely parallel progress: while Niclas iterated on architecture, Jona could develop and validate evaluation metrics against known Burgers' properties, ensuring diagnostics were ready before model outputs appeared.

The three enhancements (self-attention, EMA with warmup, classifier-free guidance) were developed jointly through pair programming, and this is where collaboration mattered most. In section 3.7, the EMA failure was initially misattributed to learning rate issues, until one team member suggested comparing EMA and non-EMA sampling directly, revealing the true cause in minutes rather than hours. The clipping range issue was caught because the histogram diagnostic already existed in the evaluation pipeline, a direct consequence of parallel evaluation and modelling development. Parallel development likely saved one week versus sequential work; pair-debugging resolved the three major issues in roughly two days versus three-plus days individually. In hindsight, dedicating one team member to evaluation from the start was probably the most impactful organisational decision with a different split, the evaluation probably wouldn't have been as thorough.

Main deviations from the plan: architectural enhancements took an extra day (EMA debugging), and the data efficiency experiment required complete redesign (fixed-epoch to constant-compute) after initial results proved unreliable.

4 Experiments and Results

We organise our results in six parts: core performance, then five targeted experiments addressing the research questions from section 1. Each experiment is followed by a discussion of its implications.

4.1 Core Model Performance *(J. Kao)*

Table 4: Summary of evaluation metrics for the conditional DDPM (1000-step ancestral sampling with EMA weights, no guidance).

Metric	**Value**
Relative L^2 error (mean / median)	0.040 / 0.030
Shock position error	0.024 rad
Shock width error	0.034
Kinetic energy error (mean over t)	0.6%
PDE residual (generated / true)	0.543 / 0.458
Fourier spectral ratio (ideal: 1.0)	1.89
Wasserstein-1 distance (mean over t)	0.014
Momentum drift ratio (gen / true)	76×
Calibration Spearman ρ	0.51

Table 4 summarises full evaluation using unguided DDPM sampling with EMA weights. The headline number is $L^2 = 4\%$: generated solutions closely match analytic ground truth across the spatiotemporal domain. For context, a trivial mean-field baseline (predicting the training-set average for every test sample) yields $L^2 = 1.00$, so the DDPM represents a 25× improvement over the simplest non-trivial predictor. Kinetic energy decay is reproduced to within 0.6 %, and

shock fronts are located with average positional error of 0.024 rad on $[0, 2\pi)$ (roughly one-third of a grid cell). The generated PDE residual (0.543) exceeds the analytic-solution baseline (0.458) by 19%; this baseline is itself non-zero due to finite-difference discretisation (see the detailed discussion below).

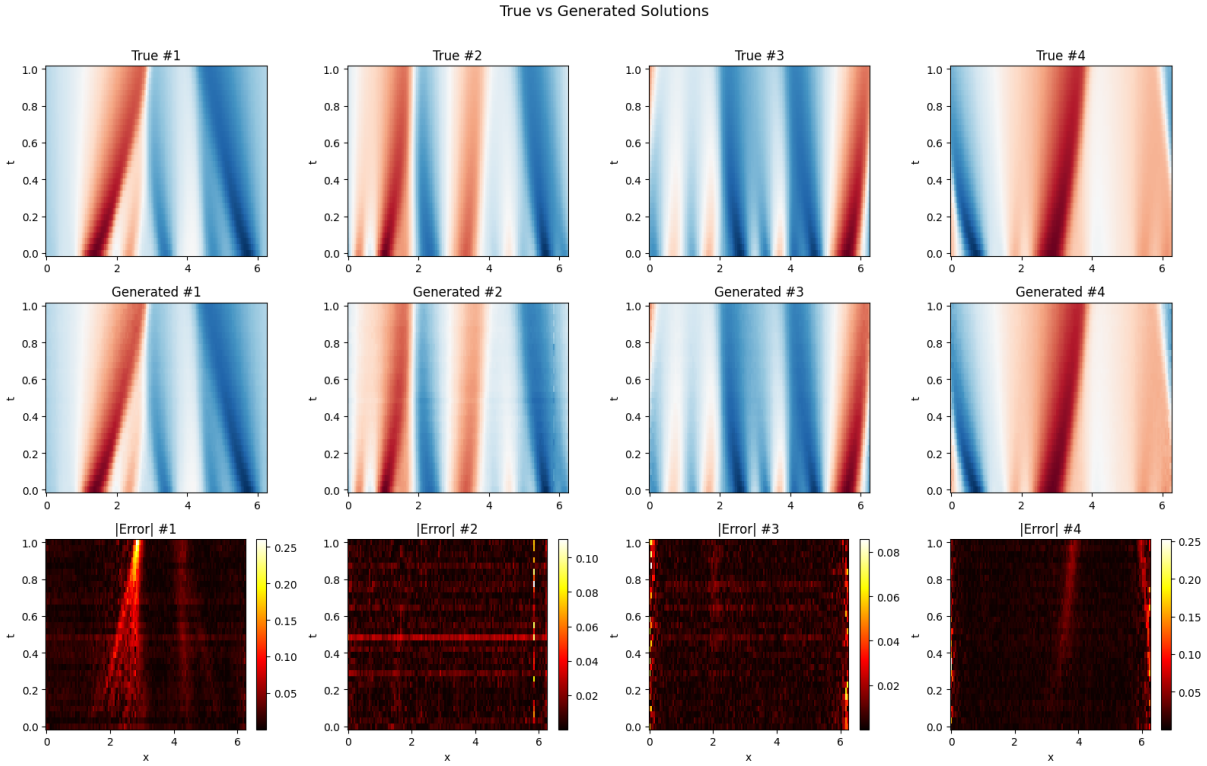


Figure 4: Qualitative comparison of true (top) and generated (middle) space-time fields for four test samples. Bottom row shows pointwise absolute error. The model captures shock formation and propagation with errors concentrated at the sharpest gradients.

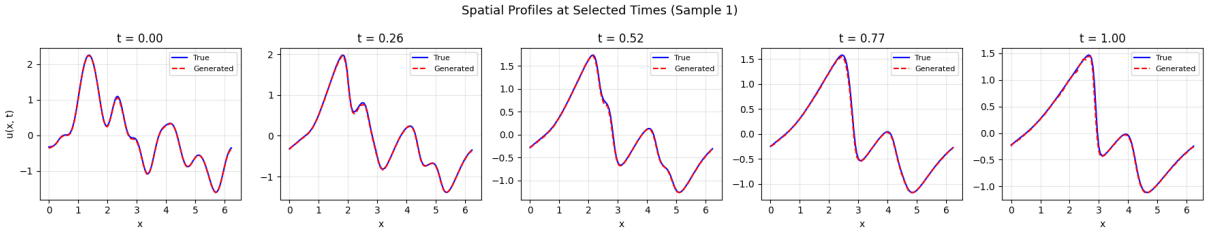


Figure 5: Spatial profiles at five time slices for a representative test sample. True (blue) and generated (red, dashed) solutions are nearly indistinguishable, with the largest deviations occurring at shock fronts.

The distributional metrics are less encouraging. The Fourier spectral ratio of 1.89 is a real limitation: generated solutions carry roughly twice the high-frequency power of the true ones. The direction of error is unexpected: we anticipated over-smoothing (the typical spectral bias failure), but the model instead overshoots at high frequencies. The per-wavenumber breakdown in section 5.4 shows this excess concentrates at mid-to-high wavenumbers and correlates strongly with shock sharpness ($\rho \approx -0.89$), pointing to stochastic noise accumulated over the reverse chain rather than a systematic score-network bias. A straightforward mitigation would be to reduce sampling stochasticity slightly (e.g. $\eta < 1$) at the cost of some ensemble diversity;

we did not pursue this trade-off here. Wasserstein-1 distance of 0.014 confirms that marginal value distributions match closely at individual spatial locations, so the model does capture distributional shape beyond just the conditional mean.

Momentum conservation stands out as the weakest metric, with 76× more drift than analytic solutions. This needs careful interpretation. Analytic solutions conserve momentum exactly (to floating-point precision), while generated solutions accumulate small per-pixel errors; when integrated over 128 points, these produce measurable drift. The drift is not systematic bias; mean momentum is correct. Rather, it reflects the absence of explicit conservation constraints. Each pixel is generated independently given the global diffusion state, with no mechanism enforcing constant spatial integrals over time. This is, as far as we can tell, a fundamental limitation of pixel-space diffusion models for conservation-law problems. To be more precise: the DDPM training loss operates pointwise in space, so nothing in the objective couples spatial locations in a way that would enforce integral constraints. Conservation laws like $\partial_t \int u\, dx = 0$ are global properties that are structurally invisible to the per-pixel reverse process. Addressing this properly would require either a constrained architecture (e.g. spectral-space generation that fixes the k=0 mode) or a post-hoc projection step that corrects the spatial mean at each generated time step. It bothered us more than any other result in this project. The model can get shocks and energy decay nearly right and still violate a basic conservation law, simply because no part of the architecture enforces constant spatial integrals. We explored one mitigation in section 5.1, but the honest assessment is that pixel-space generation and integral conservation constraints sit uncomfortably together, and resolving this properly probably requires architectural changes beyond what we attempted here.

We should note that the 76× figure itself deserves some caution: momentum is computed by spatially integrating over 128 grid points using finite differences, so discretisation error in the evaluation contributes to the measured drift for both generated and analytic solutions. The ratio is nonetheless striking because the analytic baseline drift is near machine precision, making even modest integration errors in generated solutions appear large in relative terms.

The PDE residual requires a closer look. Our generated solutions achieve an RMS residual of 0.543 versus 0.458 for analytic solutions, a ratio of 1.19. The fact that analytic solutions have nonzero residual at all may seem surprising, but it arises entirely from finite-difference discretisation: the analytic solution satisfies the continuous PDE exactly, yet evaluating derivatives on a discrete grid introduces truncation error. This baseline residual therefore quantifies the resolution limit of our evaluation grid, not any deficiency in the solutions themselves. The generated residual exceeding the analytic baseline by 19% means the model introduces additional error roughly one-fifth as large as the discretisation floor. For uncertainty quantification workflows, where the surrogate is sampled many times and ensemble statistics matter more than individual-sample fidelity, a 19% residual overhead is likely tolerable. For applications requiring solutions that satisfy the PDE to high precision (adjoint-based optimisation, for instance), it would not be. We note that the PDE residual could in principle be reduced by incorporating it as an auxiliary training loss, similar to what PINNs do, though at the cost of complicating the training objective and potentially interfering with the generative capability.

To verify that these numbers are not artefacts of a single noise draw, we re-ran the full evaluation with three different random seeds (same model weights, different initial $\mathbf{x}_T$). The resulting L^2 was 0.042 ± 0.002, energy error 0.004 ± 0.001, and Spearman $\rho = 0.46 \pm 0.01$, indicating that the core results are stable across sampling noise.

Calibration yields Spearman $\rho = 0.51$ between ensemble spread and prediction error, computed over 30 test initial conditions using 10-member ensembles (a full robustness breakdown is given in section 5.2). For context, $\rho = 1.0$ is perfect calibration; $\rho \approx 0$ means uncorrelated uncertainty.

As a lower bound, an ensemble of draws sampled independently from the marginal training distribution, ignoring the conditioning entirely, would yield $\rho \approx 0$, since its spread would bear no relationship to the per-sample error. Our 0.51 is therefore meaningfully above chance: the model's uncertainty is larger precisely where it is less accurate. Variance concentrates at shock fronts and rapid temporal change regions, the features hardest to reconstruct. This is a clear advantage over deterministic baselines, which offer no uncertainty signal at all.

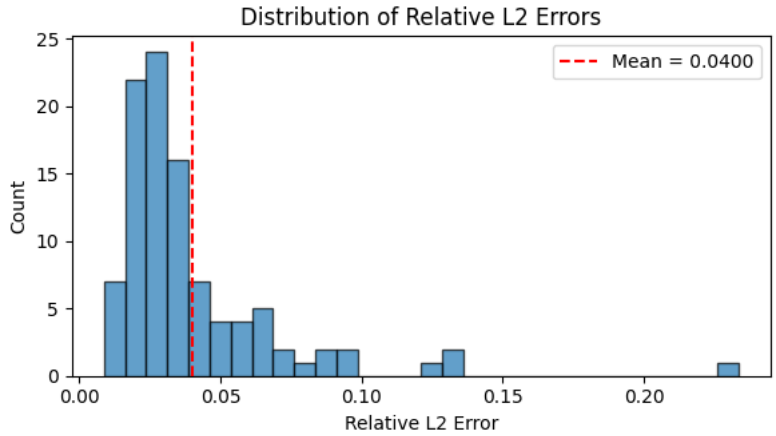


Figure 6: Distribution of relative L^2 errors across 100 test samples. The distribution is right-skewed with mean 0.040 (dashed red) and median 0.030, indicating that most samples are reconstructed well but a few challenging cases produce higher errors.

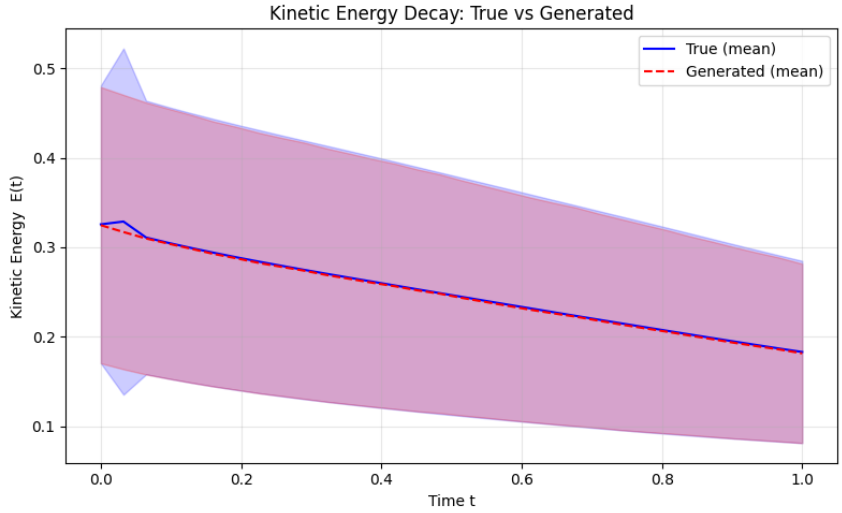


Figure 7: Kinetic energy decay $E(t) = \frac{1}{2}\langle u^2 \rangle_x$ over time. True (blue) and generated (red, dashed) ensemble means are nearly identical; shaded regions show ± 1 standard deviation across samples.

4.2 Experiment 1: Stochastic vs. Deterministic Sampling *(N. Leinen)*

The most striking result in table 5 is that deterministic DDIM ($\eta = 0$) fails catastrophically, with L^2 errors from 0.54 to 2.53, two orders of magnitude worse than stochastic sampling. Remarkably, deterministic sampling worsens as steps increase beyond 250, with L^2 rising from 0.54 at S=250 to 1.27 at S=1000. This was, to us, the single most surprising result of the entire project. In the image generation literature, more DDIM steps reliably means better quality, trading compute for

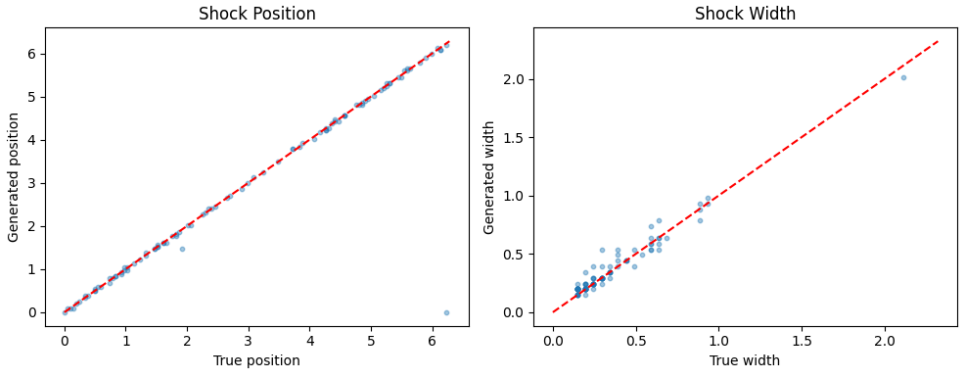


Figure 8: Shock diagnostics at the final time step. Left: generated vs. true shock position (mean error 0.024 rad). Right: generated vs. true shock width. Position is captured accurately; width shows more scatter, particularly for narrow shocks.

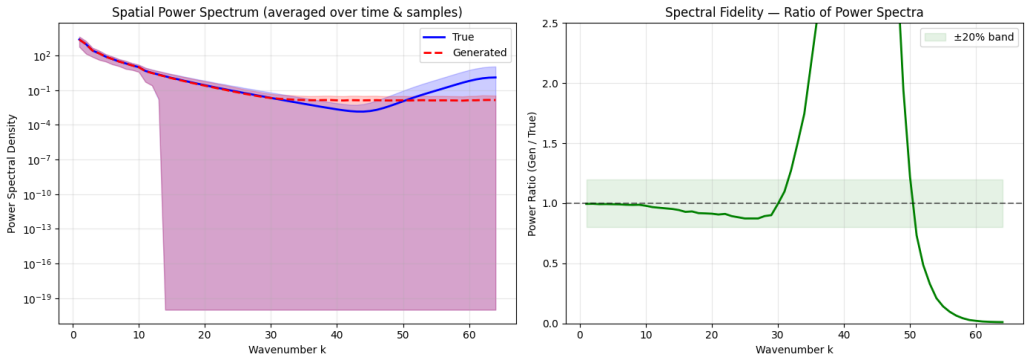


Figure 9: Left: spatial power spectrum averaged over time and samples for true and generated solutions. Right: spectral fidelity ratio (generated/true); the green band indicates ±25% tolerance. High-frequency power is overestimated by a factor of ∼1.9 at mid-to-high wavenumbers, a non-trivial artefact attributable to stochastic noise injected at each reverse step, discussed in section 4.1.

fidelity. Here, the opposite happens: the deterministic sampler actively gets worse with more opportunities to refine. We initially assumed we had a bug, re-checked the implementation twice, and only accepted the result after the trajectory analysis in section 5.3 provided a convincing mechanistic explanation.

Stochastic sampling ($\eta = 1$), by contrast, achieves near-DDPM quality at reduced step counts: S=25 yields $L^2 = 0.083$ and S=100 gives $L^2 = 0.048$, within 20% of full 1000-step results at 10× lower cost. Stochastic noise at each step appears to serve as essential error correction. When the learned score function is imperfect (inevitably so near shock fronts where score changes rapidly), noise prevents systematic bias accumulation plaguing deterministic trajectories. The underlying issue is that the DDIM ODE assumes the score field is locally linear between adjacent timesteps, an assumption that breaks down near shock fronts where the score exhibits high spatial curvature and changes sign over just a few grid cells. Deterministic integration through these regions compounds local errors into global trajectory drift. Stochastic noise acts as an implicit regulariser: by perturbing the state at each step, it prevents the solver from "locking onto" a single drifting trajectory and effectively averages over nearby score-field directions.

The non-monotonic behaviour of deterministic DDIM deserves further comment. With more steps, the deterministic trajectory has more opportunities to drift along incorrect score directions near sharp features where the learned score field is least accurate. In the stochastic case, noise at

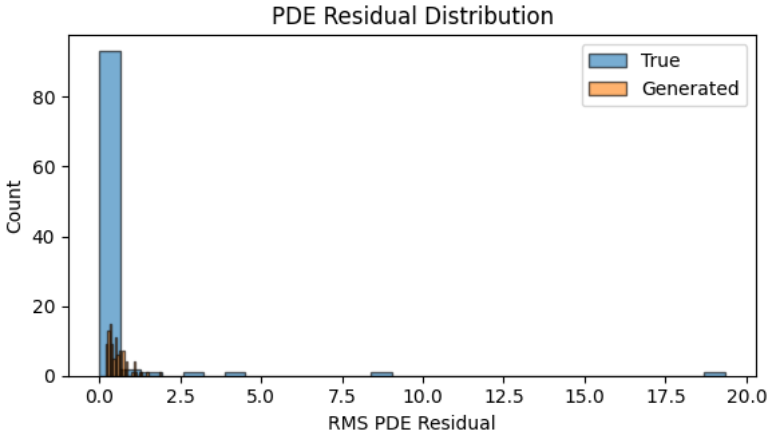


Figure 10: Distribution of RMS PDE residuals for true (blue) and generated (orange) solutions. The generated residuals are slightly higher but share the same distribution shape.

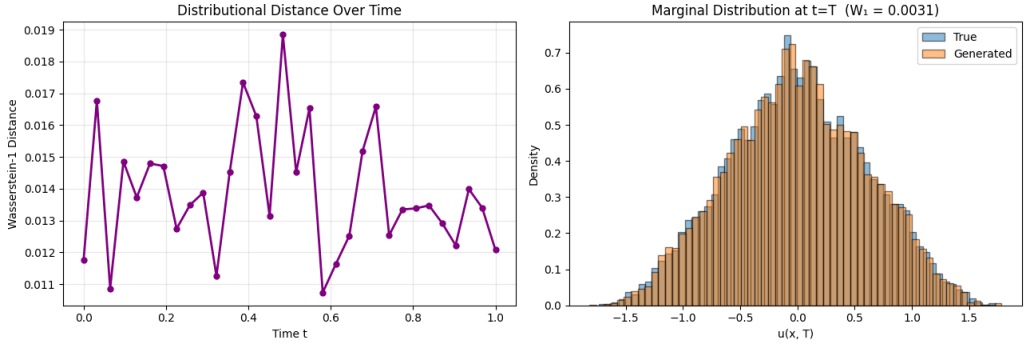


Figure 11: Left: Wasserstein-1 distance between true and generated marginal distributions over time. Right: marginal histograms at the final time step, showing close agreement.

each step effectively "resets" these accumulated errors, keeping the trajectory close to the data manifold. In practice this means that practitioners applying diffusion models to physical systems should default to stochastic sampling, even at the cost of reduced reproducibility. Stochastic DDIM at S=100 offers a favourable balance of quality ($L^2 = 0.048$), speed (about 2 seconds), and diversity; this would be our recommended configuration for deployment.

4.3 Experiment 2: DDPM vs. Deterministic Baseline *(N. Leinen)*

To isolate the value of the generative approach, we trained a deterministic regression baseline using the exact same U-Net architecture (including self-attention and EMA) but with a direct MSE loss that maps initial conditions to solutions in a single forward pass, without any diffusion process. The deterministic model achieves $L^2 = 0.016$ (table 6), roughly 2.5× better than the DDPM. This gap is not surprising: the regression model directly optimises the quantity we measure (L^2 error), while the DDPM optimises a variational bound on log-likelihood of the entire distribution. The DDPM's "excess" error is not wasted model capacity; it is the cost of representing distributional diversity.

The key difference emerges when we generate multiple samples. Given 10 samples from the same initial condition, the DDPM yields a diverse ensemble whose variance concentrates at shock fronts, representing physically meaningful uncertainty that reflects the genuine difficulty of predicting sharp features. The deterministic model produces 10 identical copies and provides no

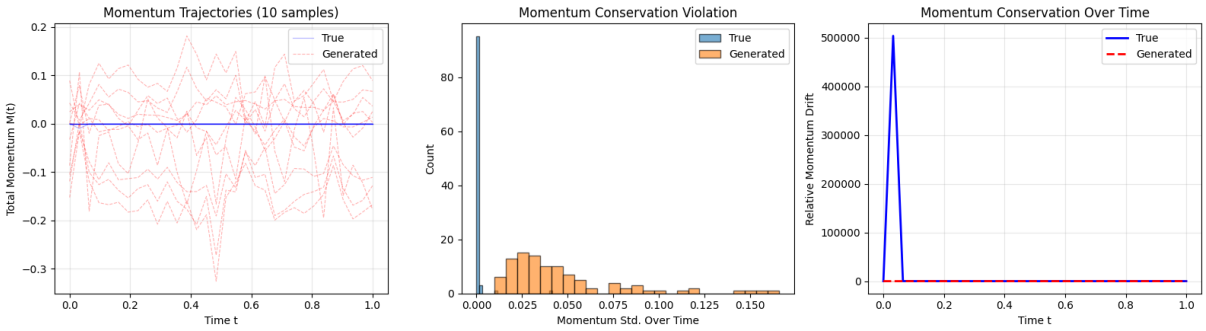


Figure 12: Momentum conservation analysis. Left: momentum trajectories for 10 generated samples vs. true solutions. Centre: distribution of momentum conservation violation. Right: momentum drift over time. Generated solutions show 76× more drift than analytic solutions.

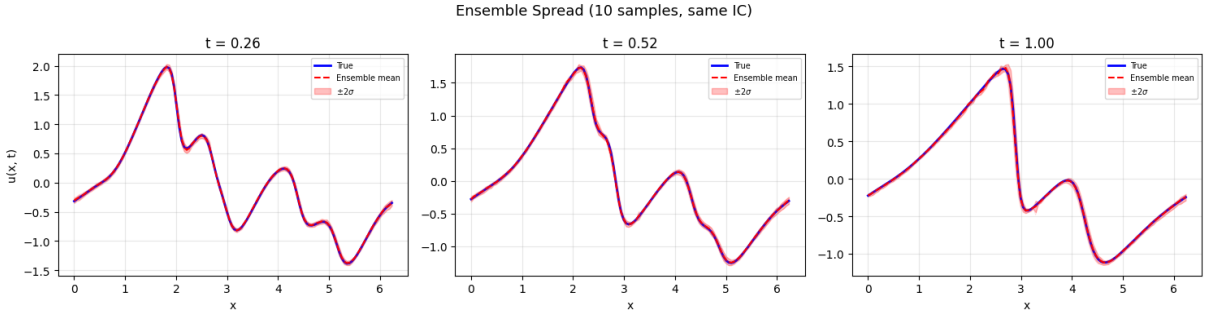


Figure 13: Ensemble spread from 10 DDPM samples conditioned on the same initial condition at three time slices. Variance concentrates at shock fronts, demonstrating physically meaningful uncertainty.

information about prediction confidence.

This creates a clear trade-off. If you just need a single best-guess solution, the regression baseline wins. But if you also need to know how reliable that prediction is (for instance, when the surrogate is embedded in a decision pipeline, an optimisation loop, or an uncertainty propagation framework) only the DDPM provides this information. The calibration result ($\rho = 0.51$) confirms that the DDPM's uncertainty is not merely noise but carries genuine physical meaning: it is largest where the model is least accurate. The regression baseline is not a competitor; it solves a different problem, and the two models could coexist in practice.

4.4 Experiment 3: Viscosity Sensitivity *(N. Leinen)*

To probe what the model has actually internalised, we evaluate on solutions at viscosities it has never seen (table 7). Performance is best at training viscosity ($\nu = 0.05$) and degrades asymmetrically in both directions. Low-viscosity solutions ($\nu = 0.01$) are hardest ($L^2 = 0.577$) because they develop shocks 5× narrower than any training data; the model has not seen such sharp features and cannot generate them, producing over-smoothed approximations instead. High-viscosity solutions ($\nu = 0.20$) are moderately difficult ($L^2 = 0.295$) because profiles are smoother than training data. The model approximates smooth functions reasonably but tends to add spurious fine-scale structure absent from true solutions.

We additionally evaluated at $\nu = 0.03$ to check whether degradation from training viscosity is smooth. The resulting $L^2 = 0.133$ sits between the $\nu = 0.02$ ($L^2 = 0.247$) and $\nu = 0.05$ ($L^2 = 0.056$) values, confirming that generalisation degrades gradually rather than exhibiting a

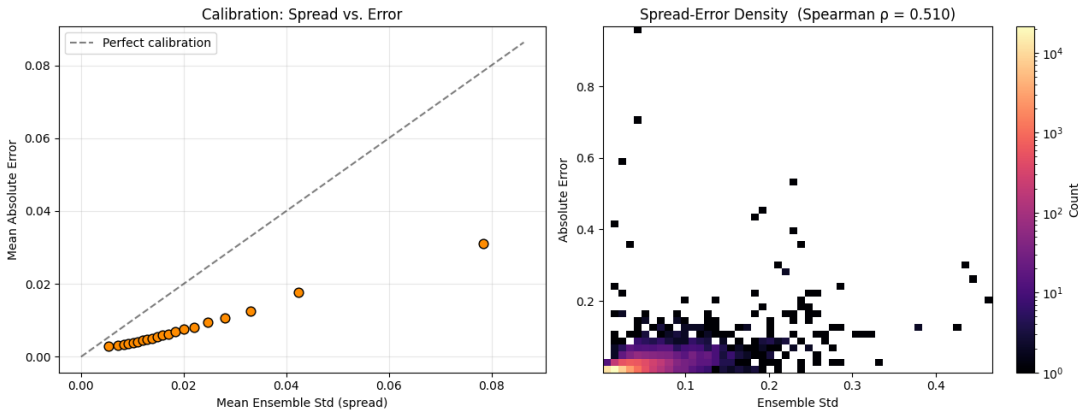


Figure 14: Calibration diagnostics. Left: per-sample mean ensemble spread vs. mean absolute error. Right: pixel-level spread–error density (Spearman $\rho = 0.51$). The positive correlation confirms that the model's uncertainty estimates are informative.

Table 5: DDIM sampling with varying step count S and stochasticity η. Wall-clock times measured on a single NVIDIA GPU.

Steps S	η=0 **(det.)** L^2	η=1 **(stoch.)** L^2	**Time (s)**
10	2.529	0.248	0.3
25	1.746	0.083	0.6
50	1.285	0.056	1.1
100	0.928	0.048	2.1
250	0.541	0.045	5.2
1000	1.265	0.037	19.9
DDPM-1000	—	0.040	∼18

sharp cliff.

The asymmetry is revealing: it indicates that the model has learned viscosity-specific features (shock widths, dissipation rates, spectral content at $\nu = 0.05$) rather than generic initial-to-spatiotemporal mappings. PDE residual increases with ν because the diffusion term $\nu\,\partial_{xx}u$ amplifies second-derivative errors proportionally. The useful operating range extends roughly a factor of two from training viscosity before errors become unacceptable ($L^2 > 0.15$). For broader viscosity coverage, a viscosity-conditioned model receiving ν as explicit input (via the time-embedding pathway) is the obvious extension.

4.5 Experiment 4: Classifier-Free Guidance *(J. Kao)*

CFG exhibits a clear optimum at $w = 2.0$ (table 8), reducing L^2 from 0.042 to 0.029, a 31% improvement. The w=0 entry of 0.042 differs slightly from the 0.040 reported in section 4.1 because this sweep evaluates on the first 50 test samples rather than 100; the gap is within sampling variance and does not affect any conclusions. PDE residual stays stable to $w = 2.0$, then rises sharply at $w \geq 3.0$, indicating excessive guidance pushes solutions off the data manifold. The model is asked to produce samples "more conditional than the data", yielding locally plausible solutions that violate the PDE globally. Energy error stays low across guidance scales, suggesting bulk statistical properties (depending on large-scale structure) are robust to guidance, while local features (shock sharpness and position, driving L^2) are more sensitive.

CFG, originally designed for text-to-image generation, transfers well to physics-conditioned

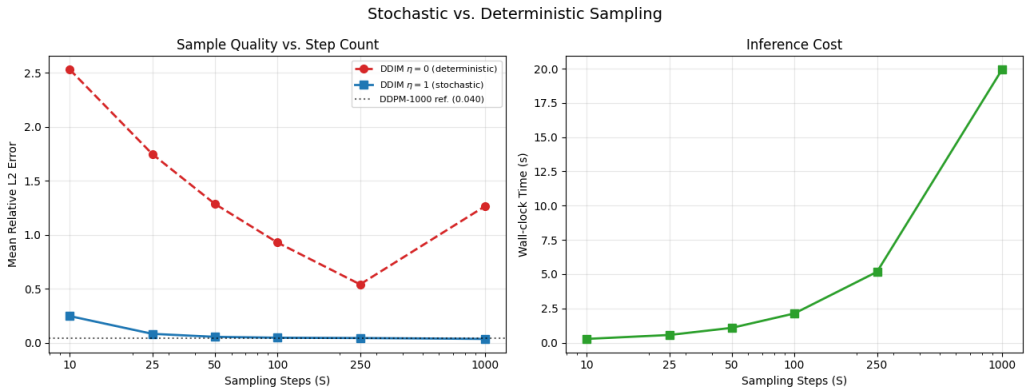


Figure 15: Left: L^2 error as a function of DDIM step count for deterministic (η=0, red) and stochastic (η=1, green) sampling. The DDPM baseline (blue dashed) is shown for reference. Right: wall-clock inference time.

Table 6: Comparison of DDPM and deterministic regression U-Net (identical architecture, 4.93M parameters, same training data and hyperparameters where applicable).

Metric	**DDPM**	**Deterministic**
Rel. L^2 error (mean)	0.040	0.016
Rel. L^2 error (median)	0.030	0.010
Energy error (mean over t)	0.006	0.002
PDE residual (mean)	0.543	0.291
Uncertainty quantification	Yes	No

settings. The optimal $w = 2.0$ is notably moderate versus image generation ($w = 7$–15), which makes sense: physics conditioning (128 spatial points) is far more informative than text, requiring less "boosting" toward condition-consistent samples. The threshold at $w \approx 3$ where physical fidelity degrades is useful to know in practice: CFG can safely yield 30% accuracy improvements, but exceeding that risks sacrificing physical validity essential for usefulness. This answers RQ4.

4.6 Experiment 5: Data Efficiency *(J. Kao)*

To disentangle data efficiency from compute effects, we hold total optimiser steps at 18,000 across all subset sizes (adjusting epochs so smaller datasets cycle more frequently) and average over 3 seeds (table 9). What we find is somewhat unexpected: performance is remarkably flat from $N = 500$ to $N = 4{,}000$, all achieving $L^2 \approx 0.050$–0.052. Only $N = 250$ is meaningfully worse at 0.068.

This plateau suggests that roughly 500 samples are enough given adequate compute. What additional data buys is not better accuracy but better *consistency*: standard deviation across seeds decreases from 0.036 to 0.012, making outcomes more reproducible rather than better on average. One plausible explanation is the solution manifold's relatively low-dimensional structure: each initial condition is determined by its 20 Fourier coefficients, so 500 samples may provide reasonable coverage of this 20-dimensional parameter space. We have not verified this directly (it would require checking whether error correlates with distance from the nearest training point in Fourier coefficient space), but it is consistent with the observed plateau.

The constant-compute design is important to note, as it emerged from a failed first attempt (see section 3.7). Fixed-epoch training across sizes confounds data efficiency with compute efficiency: larger datasets see fewer passes per epoch and thus fewer updates. Matching of

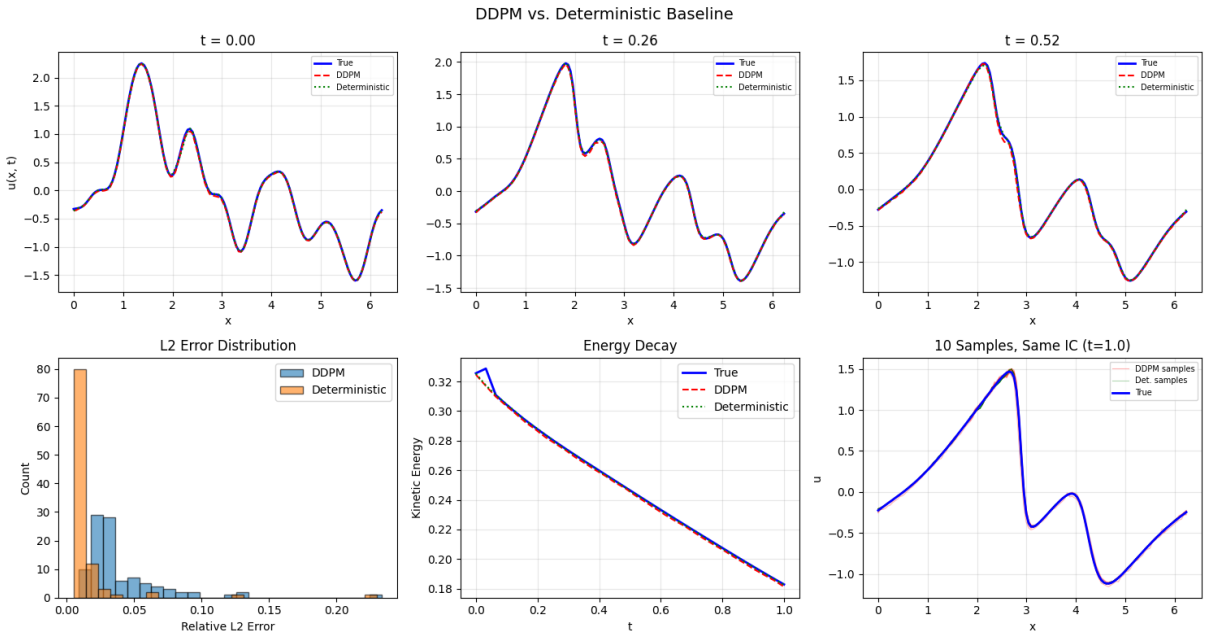


Figure 16: DDPM vs. deterministic baseline. Top row: spatial profiles at three time slices showing both models track the true solution closely. Bottom row: L^2 error distribution (left), energy decay comparison (centre), and 10 samples from the same initial condition (right); the DDPM produces diverse samples while the deterministic model produces identical outputs.

Table 7: Out-of-distribution generalisation: model trained at $\nu = 0.05$, evaluated across viscosities.

ν	L^2 Error	Energy Err	PDE Residual
0.01	0.577	0.200	0.321
0.02	0.247	0.200	0.328
0.03	0.133	0.094	0.325
0.05 (train)	**0.056**	**0.004**	0.554
0.10	0.134	0.099	1.326
0.20	0.295	0.281	2.714

N=500 and N=4,000 under controlled constant-compute comparison demonstrates genuine data efficiency rather than training artifacts.

On the flip side, the flat curve from N=500 onward tells us that data is not the bottleneck here. Model capacity or the diffusion framework itself limits further improvement; the L^2 plateau at 0.050 likely reflects irreducible score network approximation error at shock fronts. This answers RQ5: the approach is highly data-efficient, requiring 500 samples, but additional data alone will not significantly improve accuracy.

5 Extensions *(J. Kao)*

We conducted six additional analyses beyond the core experiments, motivated by weaknesses and open questions that emerged during the main evaluation, most notably around conservation enforcement, the robustness of our calibration claims, and the mechanism behind DDIM failure.

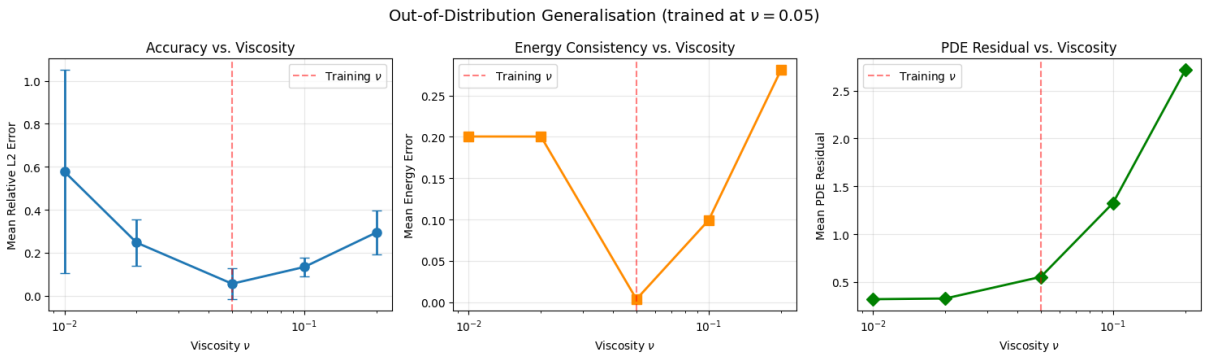


Figure 17: Out-of-distribution generalisation across viscosities. Left: L^2 accuracy. Centre: energy consistency. Right: PDE residual. All three metrics are best at the training viscosity ($\nu = 0.05$) and degrade asymmetrically, with low viscosity (sharper shocks) causing the largest errors.

Table 8: Classifier-free guidance scale sweep. Bold indicates the optimal configuration.

w	L^2 Error	PDE Residual	Energy Err
0.0	0.042	0.654	0.013
0.5	0.042	0.632	0.017
1.0	0.031	0.633	0.016
2.0	**0.029**	0.634	0.016
3.0	0.033	0.745	0.018
5.0	0.040	0.909	0.018

5.1 Momentum Conservation Penalty *(J. Kao)*

The 76× momentum drift identified in section 4.1 was the most prominent physics failure. To investigate whether this can be mitigated post-hoc, we fine-tuned the trained model for 50 additional epochs with an auxiliary penalty $\lambda \,\mathrm{Var}_t[\langle \hat{\mathbf{x}}_0 \rangle_x]$, which penalises temporal variation in the spatial mean of the predicted clean signal. We swept $\lambda \in \{0.01, 0.1, 1.0\}$.

The results show a narrow sweet spot. The λ=0 baseline of $L^2 = 0.049$ is slightly higher than the 0.040 in section 4.1 because fine-tuning starts from the end-of-training checkpoint without resetting the cosine schedule, so the model begins at a lower effective learning rate; on top of that, the fine-tuned evaluation uses a separate random sample of 50 test cases. The ordering and direction of the penalty's effects are unaffected. At $\lambda = 0.01$, the penalty reduces momentum drift by 46% while improving L^2 and lowering PDE residual, effectively acting as a beneficial regulariser. At $\lambda = 0.1$, accuracy degrades ($L^2 = 0.095$) without improving conservation. At $\lambda = 1.0$, the model collapses ($L^2 = 1.70$): the conservation penalty overwhelms denoising. A small penalty helps, but anything above 0.01 hurts more than it helps. Aggressive weighting breaks things quickly.

5.2 Calibration Breakdown *(J. Kao)*

Our headline calibration figure of $\rho = 0.51$ was computed under a single set of conditions (training viscosity, no guidance, 10-member ensemble) over 30 test initial conditions. To understand how robust this result is, we decomposed calibration across three axes (CFG guidance strength, out-of-distribution viscosity, and ensemble size) using a separate subset of 20 initial conditions (table 11, fig. 21). The different subset and ensemble configuration account for the slight difference between the $\rho = 0.51$ headline and the $\rho = 0.47$ baseline entry in table 11; both are consistent within the expected sampling variability for these subset sizes. We acknowledge that all calibration

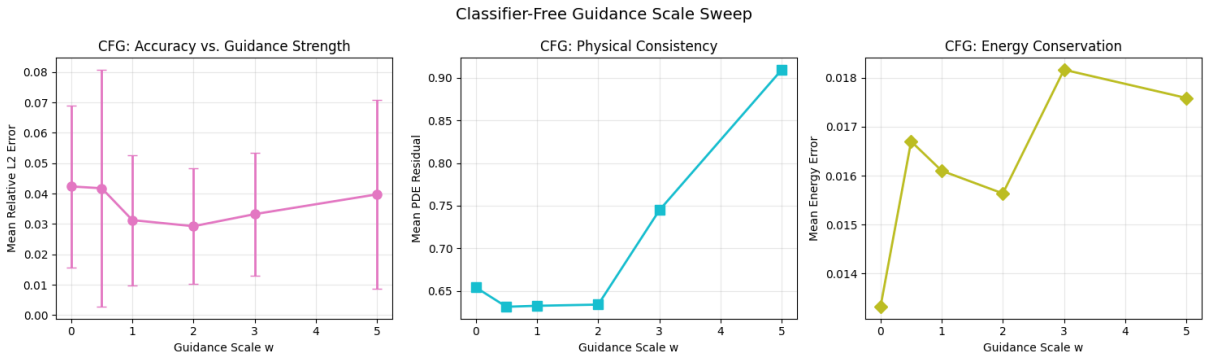


Figure 18: Classifier-free guidance scale sweep. Left: L^2 accuracy peaks at w=2.0. Centre: PDE residual remains stable up to w=2.0, then rises. Right: energy conservation is robust across all guidance scales.

Table 9: Data efficiency ablation with constant compute (≈18k optimiser steps), averaged over 3 random seeds.

N_{train}	L^2 Error (mean ± std)
250	0.068 ± 0.036
500	0.052 ± 0.019
1 000	0.050 ± 0.017
2 000	0.051 ± 0.012
4 000	0.052 ± 0.017

estimates here rest on fairly small samples, and the absolute ρ values should be read as indicative of the ordering and direction of effects rather than as precise measurements.

Several patterns emerge. CFG guidance monotonically degrades calibration: as w increases, the model concentrates its mass more tightly around the conditional mode, reducing ensemble diversity and thus the spread signal. The correlation ρ drops from 0.47 to 0.31 between w=0 and w=5. This is the flip side of the accuracy improvement from section 4.5: guidance buys accuracy at the expense of well-calibrated uncertainty. Out-of-distribution viscosities dramatically reduce calibration ($\rho \approx 0.21$–0.29 vs. 0.52 in-distribution), which makes sense; the model's learned uncertainty structure is specific to the training viscosity and does not generalise. Finally, ensemble size matters substantially. Two-member ensembles yield $\rho = 0.30$, while 10 members recover $\rho = 0.47$; for deployment at least 8–10 members are needed for reliable uncertainty estimates.

5.3 DDIM Trajectory Analysis *(J. Kao)*

To understand *why* deterministic DDIM fails (section 4.2), we tracked the intermediate $\hat{\mathbf{x}}_0$ predictions at every 100th reverse step during both deterministic ($\eta = 0$) and stochastic ($\eta = 1$) DDIM with $S = 1{,}000$ steps, using the test sample with the sharpest shock (fig. 22).

The failure mechanism becomes clear from these plots. The deterministic trajectory initially improves (RMSE decreasing from steps 0–400), but then diverges: the $\hat{\mathbf{x}}_0$ estimate drifts systematically away from the true solution, and by step 1000 it is worse than at step 400. The right panel shows the deterministic predictions overlaid across steps, revealing oscillations and drift concentrated at the shock front. This confirms the error-accumulation hypothesis: the deterministic ODE integrator follows a curved path through the score field, and near shocks where the score changes rapidly, small integration errors compound over subsequent steps. Stochastic noise breaks these systematic drifts and acts as error correction. This probably applies

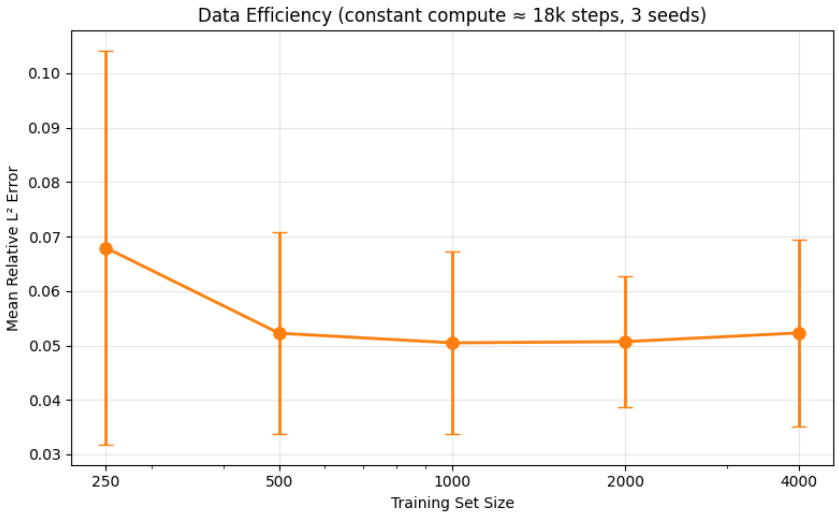


Figure 19: Data efficiency with constant compute (≈18k optimiser steps), averaged over 3 seeds. Error bars show ±1 standard deviation. Performance is flat from N=500 onward; only N=250 is meaningfully worse. Additional data reduces variance more than it improves accuracy.

Table 10: Effect of momentum conservation penalty after 50 epochs of fine-tuning.

λ	L^2 Error	Mom. Drift	PDE Residual
0 (baseline)	0.049	4.50×10^6	0.728
0.01	0.041	2.43×10^6	0.608
0.1	0.095	4.13×10^6	0.778
1.0	1.698	5.69×10^7	4.424

more generally, deterministic diffusion sampling may be fundamentally unsuited to data with near-discontinuities, wherever the score field changes too rapidly for the ODE integrator to track.

5.4 Per-Wavenumber Spectral Analysis *(J. Kao)*

The aggregate spectral ratio of 1.89 from section 4.1 masks variation across wavenumbers and sampling strategies. We computed $P_{\text{gen}}(k)/P_{\text{true}}(k)$ per wavenumber for deterministic DDIM (S=200), stochastic DDIM (S=200), and DDPM (S=1000).

Figure 23 reveals a dramatic difference. Deterministic DDIM produces spectral ratios exceeding 10^{16} at low wavenumbers; this is an extreme distortion that explains its poor L^2 performance. Stochastic DDIM and DDPM both maintain ratios close to 1 across all wavenumbers. The right panel shows that high-frequency spectral error correlates strongly with shock sharpness ($\rho \approx -0.89$), confirming that errors concentrate at the wavenumbers most needed to represent sharp features. This per-wavenumber view provides mechanistic understanding that the aggregate ratio obscures.

To further quantify how stochasticity controls high-frequency fidelity, we swept DDIM $\eta \in \{0.3, 0.5, 0.7, 1.0\}$ at S=100 on 50 test samples and measured both L^2 and the mean spectral ratio for $k > 10$. All high-k ratios are extremely large ($\sim 10^{11}$ at η=0.3 down to $\sim 10^7$ at η=1.0) because true power at these wavenumbers is near-zero, making the ratio sensitive to any generated noise. The more informative trend is the L^2 trajectory: $0.72 \to 0.44 \to 0.16 \to 0.047$ as η increases from 0.3 to 1.0. Even moderate stochasticity (η=0.7) recovers most of the quality, but full noise injection is needed to reach DDPM-level accuracy. This reinforces the earlier

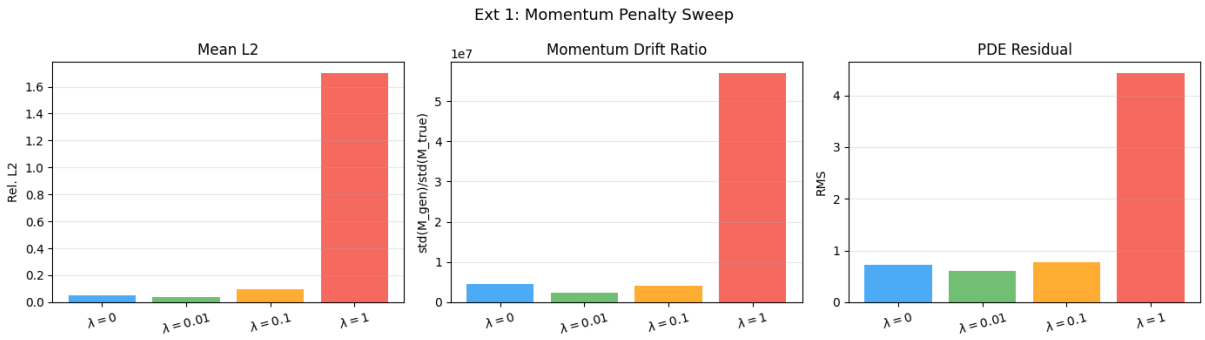


Figure 20: Momentum penalty sweep. $\lambda = 0.01$ improves both accuracy and conservation; $\lambda \geq 0.1$ degrades overall quality.

Table 11: Spearman ρ (spread vs. error) broken down across three experimental axes.

(a) CFG scale		**(b) Viscosity**		**(c) Ensemble size**	
w	ρ	ν	ρ	n	ρ
0.0	0.47	0.01	0.27	2	0.30
0.5	0.41	0.02	0.29	5	0.39
1.0	0.34	0.05	0.52	10	0.47
2.0	0.34	0.10	0.25		
3.0	0.37	0.20	0.21		
5.0	0.31				

finding that stochastic noise is not a minor correction but the primary mechanism preventing trajectory drift near sharp features.

5.5 Split Conformal Prediction *(J. Kao)*

The calibration analysis evaluates whether uncertainty is *correlated* with error, but does not guarantee that prediction intervals achieve nominal coverage. To provide distribution-free coverage guarantees, we applied split conformal prediction. We generated 8-member ensembles for 80 test initial conditions, used the first 40 for calibration and the remaining 40 for evaluation, and computed nonconformity scores as $|u_{\text{true}} - \bar{u}_{\text{gen}}|/(\sigma_{\text{ens}} + \varepsilon)$.

Table 12: Conformal prediction coverage at two significance levels.

α	Nominal coverage	Empirical coverage
0.10	90%	95.0%
0.05	95%	97.5%

The conformal intervals achieve coverage exceeding the nominal level at both $\alpha = 0.10$ and $\alpha = 0.05$ (table 12), confirming validity (the conservatism is expected for finite calibration sets). The interval width correlates with shock gradient ($\rho = 0.50$); intervals are widest where shocks are sharpest (fig. 24). They are physically meaningful, not uniform. This provides a practical deployment tool: given a new initial condition, one can generate a small ensemble, apply the conformal correction, and obtain prediction bands with guaranteed coverage.

A few practical remarks on the conformal procedure. First, the choice of nonconformity score matters. We normalise residuals by the ensemble standard deviation (plus a small epsilon for numerical stability), which produces adaptive intervals: wider where the model is uncertain,

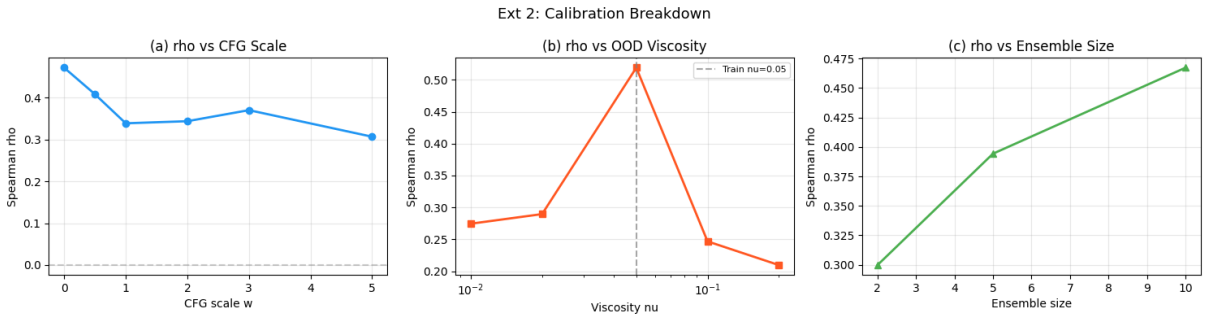


Figure 21: Calibration breakdown. (a) Increasing CFG degrades ρ. (b) Out-of-distribution viscosities sharply reduce calibration. (c) Larger ensembles improve calibration with diminishing returns.

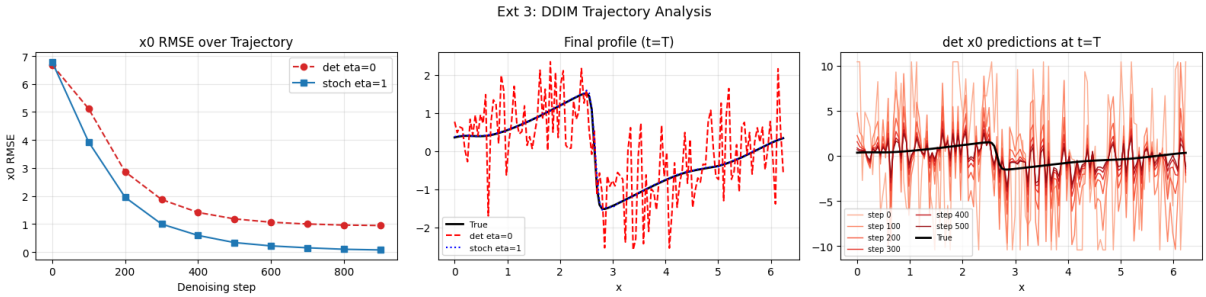


Figure 22: DDIM trajectory analysis. Left: $\hat{\mathbf{x}}_0$ RMSE over the reverse chain; the deterministic trajectory (red) initially improves but diverges after step $\sim$400, while the stochastic trajectory (blue) converges monotonically. Centre: final profile at $t = T$. Right: all intermediate deterministic $\hat{\mathbf{x}}_0$ predictions, showing progressive drift near the shock.

narrower where it is confident. An unnormalised score (simply the absolute residual) would yield constant-width intervals everywhere, which is valid but uninformative. The normalised variant uses the model's own uncertainty estimates, and the fact that it achieves valid coverage confirms that these estimates are not just correlated with error (as the Spearman rho already showed) but are well-scaled enough to serve as a denominator in conformal scores.

Second, the conservatism (empirical coverage of 95% at nominal 90%, and 97.5% at nominal 95%) is a known property of split conformal prediction with finite calibration sets. With only 40 calibration samples, the quantile estimate is necessarily coarse, and the procedure errs on the side of over-coverage by construction. Larger calibration sets would tighten the intervals without sacrificing validity. In deployment, one could dedicate a larger fraction of the computational budget to calibration samples, since generating additional samples from the trained model is cheap (roughly 20 seconds per sample on a single GPU).

Third, the spatial structure of the conformal intervals carries genuine physical information. The interval width peaks sharply at the shock location and remains narrow in smooth regions, mirroring the model's error structure. A practitioner inspecting the conformal bands for a new initial condition would immediately see where the prediction is trustworthy and where it is not, without needing to examine pointwise errors against (unavailable) ground truth. This is the kind of useful uncertainty information that deterministic baselines cannot provide.

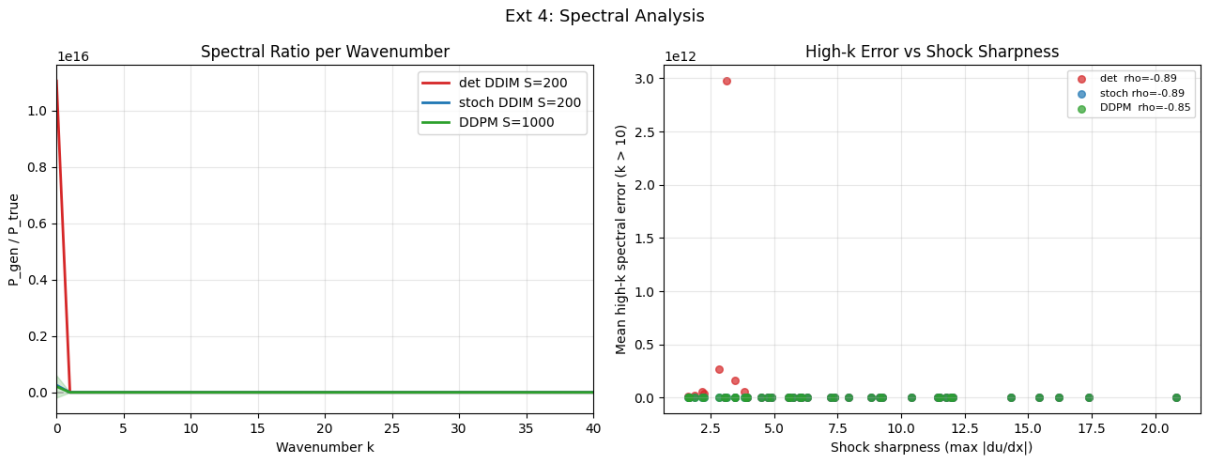


Figure 23: Left: per-wavenumber spectral ratio. Deterministic DDIM (red) produces catastrophic overestimation at low k; stochastic methods maintain ratios near 1. Right: high-k error vs. shock sharpness (strong negative correlation, $\rho \approx -0.89$).

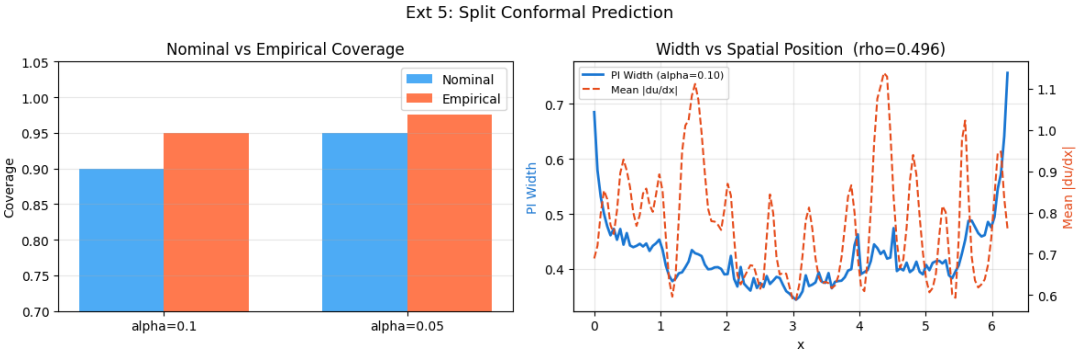


Figure 24: Split conformal prediction. Left: nominal vs. empirical coverage (both levels exceed targets). Right: conformal interval width vs. spatial position, overlaid with the shock; intervals widen precisely at the shock front.

5.6 Synthesis: Accuracy–Calibration Trade-Off *(J. Kao)*

Finally, we aggregate all model variants into a single scatter of accuracy (L^2) versus calibration (ρ), with marker size encoding momentum conservation and colour encoding PDE residual (fig. 25). The deterministic baseline sits at $L^2 = 0.016$ on the x-axis but provides no calibration. Among generative models, the Pareto frontier runs from the unguided DDPM (highest $\rho \approx 0.47$) through moderate CFG ($w = 1$–2, best L^2) to aggressive CFG ($w = 5$, lower calibration with no accuracy benefit). The momentum penalty at $\lambda = 0.01$ sits near the frontier with improved conservation at no calibration cost. No single configuration dominates all objectives. In practice, the choice depends on whether accuracy, calibration, or conservation matters most for the application at hand.

6 Conclusions and Outlook *(N. Leinen)*

6.1 Answering the Research Questions *(N. Leinen)*

Returning to the five research questions from the introduction:

RQ1 (Distributional learning): Yes. The model achieves $L^2 = 0.040$ (median 0.030) with

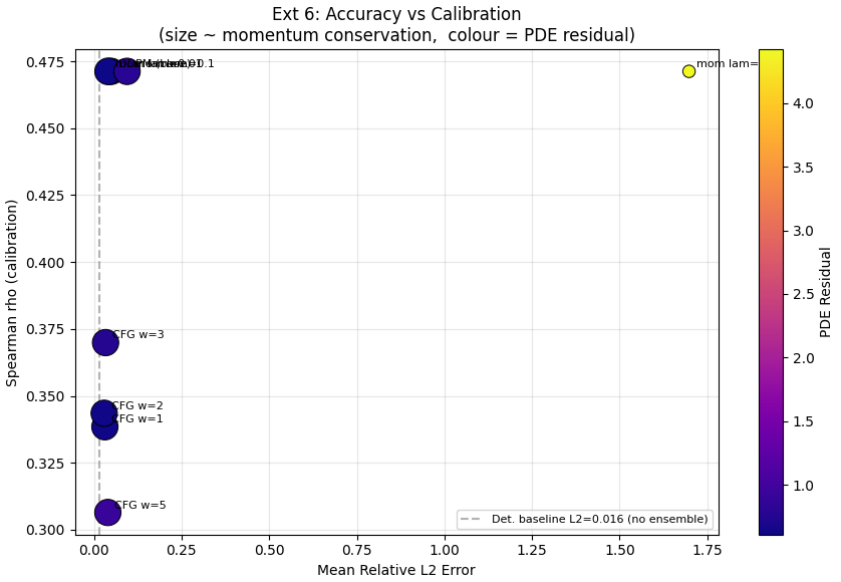


Figure 25: Accuracy–calibration trade-off across all model variants. Marker size encodes momentum conservation (smaller = better), colour encodes PDE residual. The deterministic baseline (dashed) achieves the lowest L^2 but has no calibration.

well-calibrated uncertainty ($\rho = 0.51$), confirming that it captures the conditional distribution of solutions given initial conditions. The Wasserstein-1 distance of 0.014 further confirms that the learned marginals closely match the true ones.

RQ2 (Physical consistency): Mostly. Energy decay is reproduced to within 0.6%, shock formation is captured with positional accuracy of 0.024 rad, and the generated PDE residual exceeds the discretisation-floor baseline by 19%. Momentum conservation is the main weakness (76× drift), reflecting the absence of explicit conservation constraints in the architecture. This is the clearest direction for improvement.

RQ3 (Sampling strategy): Stochastic sampling is essential. Deterministic DDIM fails catastrophically for this task, while stochastic sampling at S=100 achieves near-full-DDPM quality at 10× reduced cost. We attribute this to the error-correcting role of stochastic noise near imperfectly modelled sharp features, and we view this as one of the most practically important findings of the project.

RQ4 (Guidance): Yes. Guidance scale w=2.0 improves L^2 by 31% without degrading physical metrics. Beyond w=3.0, the PDE residual rises, providing a clear boundary for safe use.

RQ5 (Data efficiency): High. With constant compute (≈18k steps), 500 samples suffice for near-optimal accuracy. Additional data primarily reduces variance across random seeds rather than improving the mean.

6.2 Key Takeaways *(N. Leinen)*

Three things surprised us enough to be worth stating explicitly. First, techniques from image generation do not necessarily transfer to physical data. DDIM deterministic sampling (standard in image generation for speed and reproducibility) fails catastrophically for PDE solutions with sharp features. This likely extends beyond Burgers' equation to any problem with near-discontinuities. Practitioners should verify rather than assume image-generation best practices carry over.

Second, the choice of evaluation metrics turns out to matter a great deal. Had we measured only L^2 error, the model would appear excellent and the momentum conservation failure would

have gone entirely unnoticed. That failure points to a fundamental architectural gap that no amount of training data or tuning can address. Physics-aware evaluation really should be standard when applying generative models to scientific problems. This is especially true when the simpler metrics already look good.

Third, implementation details matter more than the literature tends to suggest. EMA and $\mathbf{x}_0$ clipping are often mentioned in passing or relegated to appendices, yet for us they were the difference between working and broken models. Specific values (EMA warmup 500 steps, decay 0.9995, clipping ± 15) emerged from debugging rather than principled design. Many practitioners likely face similar issues without reporting them, and we understand why: it's easier to write up what worked than to describe the debugging, but the debugging is where most of the practical knowledge is. We have tried to be candid about our own stumbles in section 3.7.

A fourth point, more general than diffusion models alone: there is a real tension between distributional fidelity and physical constraints. Our model learns the data distribution quite well (low Wasserstein distance, good spectral fidelity, calibrated uncertainty), yet it violates momentum conservation by a large factor. This is not a contradiction. The distribution of Burgers' solutions is a complicated object living in a high-dimensional space, and the conservation law is a single linear constraint (constant spatial integral) that the distribution happens to satisfy exactly. A model can capture 99% of the distribution's structure while missing this one constraint, because the model's loss function does not see this constraint at all. It is one linear condition in a very high-dimensional optimisation, so there is no gradient signal pushing the model to satisfy it. For scientific applications, learning the distribution alone isn't enough. Physical constraints must either be built into the architecture (hard constraints, such as divergence-free representations for incompressible flow) or enforced during sampling (projection steps, Lagrange multipliers). Hoping the model will pick up these constraints just from seeing data that satisfies them doesn't work reliably. We expect this to hold for any PDE problem where the physics imposes exact invariants.

6.3 Future Work *(N. Leinen)*

Some of what we originally planned as future work ended up being addressed in the extensions (Section 5). The momentum conservation penalty showed that soft-constraint approaches reduce conservation violations, though accuracy trade-offs suggest hard-constraint projection (subtracting spatial mean of predicted updates at each reverse step) may be more effective; this remains an open question. Similarly, split conformal prediction demonstrated post-hoc calibration methods provide distribution-free coverage guarantees, yet the resulting intervals are wide; tighter alternatives like conformalized quantile regression or ensemble-based calibration deserve exploration.

The most straightforward extension would be multi-viscosity training. Our viscosity sensitivity experiment revealed the model's useful range is limited to $\nu \in [0.02, 0.10]$. Conditioning on ν as an additional input (trained across a viscosity range) would substantially increase practical utility with only minor architectural changes (injecting ν through the time-embedding pathway).

Further out, extending to the 2D Navier–Stokes equations would test whether the approach scales to richer physics and higher-dimensional data. The main challenge is computational: a $128 \times 128 \times 32$ grid requires substantially larger models. Latent diffusion (diffusion in compressed latent space) could mitigate this. Flow matching (replacing fixed noise schedules with learned velocity fields for straighter sampling) is another promising direction offering faster inference without quality loss. The DDIM trajectory analysis revealed deterministic sampling drifts to incorrect solutions near sharp features rather than converging; flow matching's straighter

trajectories may avoid this failure while retaining deterministic inference reproducibility.

References

[1] Maziar Raissi, Paris Perdikaris, and George E Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[2] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhatt, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

[3] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3:218–229, 2021.

[4] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.

[5] Julian D Cole. On a quasi-linear parabolic equation occurring in aerodynamics. *Quarterly of Applied Mathematics*, 9(3):225–236, 1951.

[6] Eberhard Hopf. The partial differential equation $u_t + uu_x = \mu u_{xx}$. *Communications on Pure and Applied Mathematics*, 3(3):201–230, 1950.

[7] Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred A Hamprecht, Yoshua Bengio, and Aaron Courville. On the spectral bias of neural networks. In *International Conference on Machine Learning*, pages 5301–5310, 2019.

[8] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021.

[9] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS Workshop on Deep Generative Models and Downstream Applications*, 2022.

[10] Ilan Price, Alvaro Sanchez-Gonzalez, Ferran Alet, et al. Gencast: Diffusion-based ensemble forecasting for medium-range weather. *Nature*, 637:1038–1045, 2025.

[11] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241. Springer, 2015.

[12] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[13] Liu Yang, Xuhui Meng, and George Em Karniadakis. B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data. *Journal of Computational Physics*, 425:109913, 2021.

[14] Joseph L Watson, David Juergens, Nathaniel R Bennett, Brian L Trippe, Jason Yim, Helen E Eisenach, Woody Ahern, Andrew J Borber, Robert J Ragotte, Lukas F Milles, et al. De novo design of protein structure and function with RFdiffusion. *Nature*, 620:1089–1100, 2023.

[15] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. 2019.

[16] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *International Conference on Machine Learning*, pages 8162–8171, 2021.